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PART 2: POE

LOGISTIC REGRESSION

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1. INTRODUCTION

The publishing industry produces large amounts of data relating to books, authors, pricing, publishers and sales performance. Analysing these characteristics can help determine whether patterns exist that distinguish commercially successful books from those with lower sales. Classification techniques are particularly useful when the outcome of interest consists of two categories. Logistic Regression is a supervised statistical learning method that estimates the probability of a binary outcome based on a set of predictor variables (James et al., 2023).

The purpose of this analysis was to develop a **Logistic Regression classification algorithm from scratch using Python and NumPy** to investigate whether available book characteristics could be used to predict bestseller status. Rather than relying on a pre-built Logistic Regression estimator, the algorithm was manually developed by implementing the linear model, sigmoid function, binary cross-entropy loss, gradient calculations and gradient-descent optimisation. This approach provided greater insight into the mathematical and computational processes underlying Logistic Regression.

The analysis used a dataset containing **1,070 book records** with numerical and categorical characteristics relating to publication, authors, genre, language, publisher, pricing, ratings and sales. Because the original dataset did not contain an official binary bestseller variable, bestseller status was operationally defined according to observed unit sales. Books with unit sales above the **75th percentile of 5,312.25 units** were classified as bestsellers, resulting in **268 bestseller and 802 non-bestseller observations**.

The analytical process included dataset-quality assessment, data cleaning, exploratory data analysis, feature selection, preprocessing, model training, validation-based model improvement and final evaluation on a previously unseen test set. Particular attention was given to preventing target leakage, managing class imbalance and evaluating the model using metrics appropriate for an imbalanced binary classification problem. The final objective was not only to assess predictive performance but also to interpret the model's strengths, weaknesses and potential usefulness within a publishing context.

2. DATASET SELECTION AND SUITABILITY

2.1 Dataset Source and Description

The dataset selected for this analysis was the Books Sales and Ratings dataset obtained from Kaggle. It was published under the Kaggle username The Devastator and contains information relating to book publication, authors, ratings, genres, publishers, pricing and commercial sales performance (The Devastator, 2023). The file used for the analysis was Books_Data_Clean.csv.

The imported dataset contained **1,070 observations and 15 variables**. These included numerical variables such as *Publishing Year, Book_average_rating, Book_ratings_count*, *gross sales*, *publisher revenue*, *sale price*, *sales rank* and *units sold*. It also contained categorical information including *Book Name, Author*, *language_code*, *Author_Rating*, *genre* and *Publisher*.

The combination of numerical and categorical characteristics made the dataset suitable for investigating a binary classification problem. Logistic Regression can accommodate numerical predictors together with appropriately encoded categorical variables and is commonly used to estimate the probability of membership in one of two outcome classes (James et al., 2023).

```
Dataset loaded successfully.
Number of rows: 1,070
Number of columns: 15
```

[2]:

index	Publishing Year	Book Name	Author	language_code	Author_Rating	Book_average_rating	Book_ratings_count	genre	gross sales	publisher revenue	sale price	sales rank	Publi
0	1975.0	Beowulf	Unknown, Seamus Heaney	en-US	Novice	3.42	155903	genre fiction	34160.0	20496.0	4.88	1	HarperC Publi
1	1987.0	Batman: Year One	Frank Miller, David Mazzucchelli, Richmond Lew...	eng	Intermediate	4.23	145267	genre fiction	12437.5	7462.5	1.99	2	HarperC Publi
2	2015.0	Go Set a Watchman	Harper Lee	eng	Novice	3.31	138669	genre fiction	47795.0	28677.0	8.69	3	Am D Services
3	2008.0	When You Are Engulfed in Flames	David Sedaris	en-US	Intermediate	4.04	150898	fiction	41250.0	24750.0	7.50	3	Hacl Book G
4	2011.0	Daughter of Smoke & Bone	Laini Taylor	eng	Intermediate	4.04	198283	genre fiction	37952.5	22771.5	7.99	4	Pen Group (

Figure 1: Initial structure of the Books Sales and Ratings dataset

2.2 Dataset Quality Assessment

Before the dataset was accepted for modelling, its quality was evaluated by examining data types, missing values, duplicate observations, numerical ranges and categorical cardinality. Data quality and preprocessing are important because inconsistencies, missing observations and unsuitable predictors can negatively affect the reliability of a predictive model (Kuhn and Johnson, 2013).

The duplicate-record audit identified **no completely duplicated observations**, meaning that no records had to be removed because of exact duplication.

A limited amount of missing information was identified. language_code contained **53 missing values (4.95%)**, Book Name contained **23 missing values (2.15%)**, and Publishing Year contained only **one missing value (0.09%)**. The remaining variables contained no missing observations.

These missing values were considered manageable. Book Name was not intended to be used as a model predictor because individual titles act mainly as identifiers and have extremely high cardinality. Missing language information could be represented using an explicit Unknown category, while the single missing publication year could later be imputed using information derived from the training partition only. This approach avoids discarding a large number of otherwise usable observations.

MISSING-VALUE SUMMARY

	Missing Values	Missing Percentage
language_code	53	4.95
Book Name	23	2.15
Publishing Year	1	0.09
index	0	0.00
Author	0	0.00
Author_Rating	0	0.00
Book_average_rating	0	0.00
Book_ratings_count	0	0.00
genre	0	0.00
gross sales	0	0.00
publisher revenue	0	0.00
sale price	0	0.00
sales rank	0	0.00
Publisher	0	0.00
units sold	0	0.00

Figure 2: Missing-value summary

```
DUPLICATE AUDIT
----------------------------------------------
Completely duplicated rows: 0
Percentage of duplicated rows: 0.00%
```

Figure 3: Duplicate-record assessment

2.3 Assessment of Potential Data-Quality Problems

The numerical audit identified unusually old values in Publishing Year, including negative years associated with ancient works. These observations were investigated rather than being automatically deleted as outliers. The inspection showed that many represented genuine historical or ancient texts rather than obvious data-entry errors. They were therefore retained to avoid removing legitimate observations simply because their values were unusual.

The dataset also contained very high-cardinality categorical variables. *Book Name* contained more than one thousand distinct values and *Author* contained several hundred authors. Directly one-hot encoding these variables would have created a large number of sparse predictors relative to the 1,070 observations. High-dimensional predictor spaces can

increase model complexity and increase the risk of fitting patterns that do not generalise well to unseen data (Kuhn and Johnson, 2013). These variables were therefore not selected as direct predictors.

Another important consideration was that several variables reflected **realised commercial performance**, including *units sold*, *gross sales*, *publisher revenue* and *sales rank*. If a variable directly reveals information about the outcome that the model is intended to predict, its inclusion can create target leakage and give an unrealistically optimistic impression of predictive performance. Consequently, these realised sales variables were excluded from the predictor set once the target had been constructed.

2.4 Construction of the Bestseller Target

A major limitation of the source dataset was that it did not contain an official binary variable identifying whether each book was a bestseller. A binary target therefore had to be constructed from the available sales information.

The units sold variable was selected for this purpose because it provided a direct quantitative measurement of book sales. The 75th percentile of units sold was **5,312.25 units**. Books with sales above this value were classified as Bestseller = 1, while books at or below the threshold were classified as Bestseller = 0. The 75th-percentile value was confirmed directly from the dataset's numerical summary.

This definition produced:

Target class	Number of books	Percentage
Non-Bestseller	802	74.95%
Bestseller	268	25.05%

Table 1: Distribution of the operational bestseller target

The resulting target therefore created a meaningful but moderately imbalanced binary classification problem. Logistic Regression is appropriate for modelling a binary outcome because it estimates class probabilities through the logistic function rather than producing unrestricted continuous predictions (James et al., 2023).

It is important to note that the term **bestseller** in this analysis is an operational definition specific to this dataset. It does not indicate membership in an external bestseller list such as a national publishing ranking. Instead, it identifies books belonging to the highest-selling quarter of the available observations. This distinction is retained throughout the analysis to avoid overstating what the target variable represents.

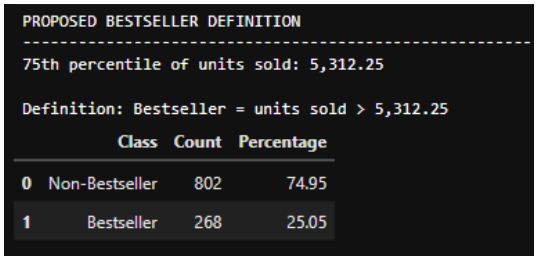


Figure 4: Operational definition and class distribution of bestseller status

2.5 Overall Suitability for Logistic Regression

Following the audit, the dataset was considered suitable for the proposed Logistic Regression analysis for several reasons. It provided more than one thousand observations, contained a mixture of potentially informative numerical and categorical variables, had no complete duplicate records and contained only a limited amount of missing metadata. The dataset also supported the creation of a clear binary target with sufficient examples in both classes.

However, suitability did not imply that the dataset was without limitations. The absence of an official bestseller label required an operational target definition, some categorical groups contained relatively few observations, and a number of highly predictive sales variables could not legitimately be used because they would leak information about realised sales performance. These issues were therefore considered explicitly during feature selection, model training and interpretation.

Overall, the dataset provided an appropriate foundation for demonstrating the full Logistic Regression workflow while also presenting realistic analytical challenges involving missing data, high-cardinality variables, class imbalance and leakage prevention.

2.6 Section Summary

The dataset was considered suitable for Logistic Regression after evaluating its structure, completeness, duplication, categorical complexity and sales-related variables. Although minor missing values and some unusual historical publication years were identified, these issues were manageable and did not prevent modelling. Bestseller status was operationally defined using the 75th percentile of unit sales, producing a binary target with sufficient observations in both classes. Potential target-leakage variables were identified at this stage and reserved from later predictor selection. Overall, the dataset provided an appropriate basis for developing and evaluating a from-scratch Logistic Regression classifier.

3. ANALYSIS PLAN

A structured analysis plan was developed before model training to ensure that the Logistic Regression workflow followed a logical sequence from data exploration through to final model evaluation. The plan was informed by the preliminary dataset audit and was designed to reduce methodological problems such as target leakage, overfitting and inappropriate use of the test dataset.

A clearly defined modelling workflow is important because predictive performance should be evaluated on observations that were not used to fit or tune the model. Separating model development from final evaluation provides a more realistic indication of how the model may perform on unseen data (James et al., 2023). The planned workflow therefore separated exploratory analysis, feature selection, preprocessing, model training, validation-based improvement and final test evaluation.

3.1 Planned Analytical Workflow

The analysis was organised into eight main stages.

Stage	Planned approach
Exploratory Data Analysis	Examine target balance, numerical distributions, categorical distributions, missing values, unusual observations and relationships between candidate predictors and bestseller status.
Feature Selection	Remove identifiers, high-cardinality variables and variables that could introduce target or temporal leakage. Statistical association tests would be used to support feature screening.
Data Preparation	Divide the dataset into stratified training, validation and test partitions. Missing numerical values would be imputed using training data only, numerical predictors standardised and categorical predictors one-hot encoded.
Model Training	Develop Logistic Regression manually using NumPy by implementing the linear model, sigmoid function, binary cross-entropy loss, gradient calculations and batch gradient descent.
Hyperparameter Definition	Begin with predetermined values for learning rate, number of iterations and classification threshold, and only modify them if validation evidence indicated a need for improvement.
Model Evaluation	Evaluate the classifier using accuracy, balanced accuracy, precision, recall, specificity, F1-score, confusion matrix and ROC-AUC.
Model Improvement	Compare alternative classification thresholds, class weighting, early stopping and feature sets using validation performance only.
Final Evaluation and Reporting	Select the final model using validation results, evaluate it once on the untouched test set and report its strengths, weaknesses, limitations and practical interpretation.

Table 2: Planned analytical workflow for the Logistic Regression bestseller-classification analysis

3.2 Exploratory Data Analysis Plan

The first analytical stage was designed to establish how the target and potential predictor variables behaved before any model was trained. This included assessing target-class balance, numerical distributions, categorical frequencies, missing values and unusual observations.

Exploratory Data Analysis is useful because it allows the analyst to understand the structure and quality of the data before selecting modelling procedures. It can also identify variables requiring transformation, encoding, exclusion or further investigation (Kuhn and Johnson, 2013).

Particular attention was planned for the binary target because an imbalanced outcome can make ordinary accuracy misleading. For example, a classifier may achieve apparently high accuracy by favouring the majority class while performing poorly on the minority class. Consequently, class-specific measures such as precision and recall were included in the evaluation strategy from the outset.

3.3 Feature-Selection Plan

Feature selection was planned using both **methodological reasoning** and **statistical evidence**.

The first stage involved identifying variables that should not be used as predictors regardless of their statistical association with the target. This included identifiers, high-cardinality variables and realised commercial-performance variables that could reveal information about bestseller status.

The second stage involved evaluating the remaining predictors using univariate statistical association tests. Categorical predictors would be assessed using chi-square tests and effect-size measures, while numerical predictors would be examined using appropriate association tests.

However, statistical significance was not intended to be the sole criterion for deleting predictors. A variable that is weak when examined independently may still contribute useful information when combined with other predictors. For this reason, the full leakage-controlled predictor set would initially be retained and later compared against a reduced feature set using validation performance.

Conventional stepwise feature-selection approaches were also considered. Forward selection begins with no predictors and adds variables sequentially according to a chosen criterion, while backward elimination begins with the full predictor set and removes variables sequentially. In this analysis, p-values from the univariate tests were used as supporting screening evidence rather than as automatic deletion rules. Because categorical variables expand into groups of dummy variables after one-hot encoding, removing individual dummy coefficients could fragment a conceptually single predictor. For this reason, the practical feature-selection experiment compared the full leakage-safe predictor set with a reduced Author_Rating model using validation performance (James et al., 2023; Kuhn and Johnson, 2013).

3.4 Data Preparation Plan

After feature selection, the dataset would be divided into separate training, validation and test partitions using stratification to preserve approximately the same bestseller proportion across each subset.

The intended roles of the datasets were:

- the **training set** would be used to estimate model parameters;
- the **validation set** would be used to compare model configurations and make improvement decisions; and
- the **test set** would remain untouched until the final model had been selected.

This separation is important because repeatedly evaluating alternative models on the test set can indirectly cause the model-development process to become adapted to that test data, leading to an overly optimistic estimate of generalisation performance (James et al., 2023).

Preprocessing operations would also be fitted using the training data only. Numerical variables would be imputed where necessary and standardised, while categorical variables would be converted to numerical representations using one-hot encoding. Standardisation was particularly relevant because the Logistic Regression algorithm would be optimised using gradient descent, and large differences in feature scale can negatively affect optimisation behaviour (Kuhn and Johnson, 2013).

3.5 Logistic Regression Training Plan

The assignment required Logistic Regression to be developed from scratch rather than using a pre-built classification estimator. The planned implementation therefore included the core mathematical components of the algorithm.

For each observation, the linear model would first calculate:

$$[z = X\omega + b]$$

where (X) represents the predictor variables, (w) represents the learned coefficients and (b) represents the intercept.

The sigmoid function would then convert the unrestricted linear output into a probability:

Logistic Regression is appropriate for binary classification because this transformation constrains predicted values to the interval between 0 and 1, allowing them to be interpreted as estimated class probabilities (James et al., 2023).

Binary cross-entropy would be used as the loss function, while gradient descent would iteratively update the weights and intercept in an attempt to minimise that loss.

3.6 Hyperparameter and Model-Improvement Plan

The initial model would use predetermined baseline settings rather than immediately optimising multiple parameters. This was intended to establish a reference model against which subsequent improvements could be compared.

The main development decisions would include:

- learning rate;
- number of gradient-descent iterations;
- classification probability threshold;

- class weighting;
- early stopping; and
- predictor-set selection.

Model changes would only be introduced when validation results provided evidence that the baseline configuration was inadequate. The final test set would not be used to determine any of these settings.

This approach allowed each modification to be treated as a controlled experiment rather than repeatedly changing the model without a clear analytical justification.

3.7 Evaluation Plan

Multiple evaluation measures were selected because no single metric provides a complete assessment of classification performance.

Accuracy would measure the overall proportion of correct predictions, but because bestseller observations represented the minority class, additional measures were required. Precision would measure the proportion of predicted bestsellers that were actually bestsellers, while recall would measure the proportion of actual bestsellers successfully detected.

The F1-score would provide a combined measure of precision and recall and was planned as an important metric for model comparison. Balanced accuracy would also be considered because it gives equal importance to both classes.

The confusion matrix would be used to identify true positives, true negatives, false positives and false negatives. ROC-AUC would provide a threshold-independent measure of the model's ability to rank bestseller observations above non-bestsellers (James et al., 2023).

3.8 Section Summary

The analysis plan established a controlled workflow for developing the Logistic Regression classifier. Exploratory analysis and feature screening were planned before modelling, while preprocessing was restricted to information obtained from the training data. The modelling strategy separated training, validation and test data so that model-improvement decisions could be made without contaminating the final evaluation. Logistic Regression was planned for manual implementation using NumPy, with subsequent improvements based on validation evidence rather than arbitrary parameter changes. Multiple evaluation metrics were selected to account for the imbalance between bestseller and non-bestseller observations.

4. DATA CLEANING AND TARGET PREPARATION

Following the preliminary dataset audit, a structured cleaning process was applied before modelling. The purpose of this stage was not to remove unusual observations automatically, but rather to correct clearly inconsistent labels, eliminate non-predictive identifiers and prepare variables in a form suitable for later analysis. Good preprocessing is important because modelling results can be affected by duplicated information, inconsistent categories and variables that do not represent meaningful predictive characteristics (Kuhn and Johnson, 2013).

A separate working copy of the raw dataset was first created so that the original imported data remained unchanged. The working dataset initially retained all **1,070 observations and 15 variables**.

4.1 Removal of the Artificial Index Variable

The index variable was removed because it merely represented the sequential row number of each observation. It did not describe any characteristic of the book and therefore had no meaningful relationship with bestseller performance.

Including such an identifier could allow the model to learn accidental patterns related to row ordering rather than genuine relationships between book characteristics and the target. Removing it reduced the dataset from 15 to 14 original variables before the newly engineered variables were added.

4.2 Investigation of Repeated Book Titles

Although the earlier audit identified no completely duplicated records, repeated book titles were investigated separately. The analysis identified **four observations involving repeated titles**.

The repeated titles involved Persepolis and The Awakening. Inspection showed that the associated observations differed in characteristics such as author, publication year, publisher, price and sales performance. They were therefore interpreted as different editions, different commercial records or distinct books sharing the same title rather than duplicate observations.

Removing these records simply because their titles matched would have resulted in the loss of legitimate information. They were consequently retained in the cleaned dataset.

Records involving repeated book titles: 4

	Book Name	Author	Publishing Year	language_code	Publisher	sale price	units sold
297	Persepolis	Marjane Satrapi, Mattias Ripa	2000.0	eng	Random House LLC	4.99	298
933	Persepolis	Marjane Satrapi, Mattias Ripa, Blake Ferris, A...	2003.0	en-US	Amazon Digital Services, Inc.	0.99	4320
251	The Awakening	Kate Chopin	1899.0	eng	Amazon Digital Services, Inc.	0.99	375
696	The Awakening	Kelley Armstrong	2009.0	eng	Simon and Schuster Digital Sales Inc	6.45	36072

Figure 5: Investigation of repeated book-title records before modelling

4.3 Standardisation of Categorical Variables

Categorical labels were examined for inconsistencies that could cause logically equivalent categories to be treated as separate predictors during one-hot encoding.

4.3.1 Genre

The source dataset contained both *genre fiction* and *fiction*. These categories represented the same broad genre and were therefore consolidated under the label *fiction*.

After standardisation, the genre distribution consisted of:

- 884 fiction books;
- 171 nonfiction books; and
- 15 children's books.

This resulted in fiction representing **82.62%** of observations.

This standardisation prevented the later encoding process from creating separate dummy variables for categories that had the same conceptual meaning.

4.3.2 Publisher

The publisher audit also identified two highly similar labels: *HarperCollins Publishing* and *HarperCollins Publishers*. These were consolidated under *HarperCollins Publishers*.

Standardising publisher names reduced artificial category fragmentation and ensured that observations from the same publishing organisation were represented consistently during modelling.

4.3.3 Language

The original *language_code* variable contained several regional codes for English, including *eng*, *en-US*, *en-GB* and *en-CA*. These were grouped into a single *English* category.

Languages represented by only a very small number of observations were grouped into *Other*, while missing language information was represented as Unknown rather than causing otherwise complete observations to be removed.

The resulting language distribution was:

Language group	Count	Percentage
English	1,008	94.21%
Unknown	53	4.95%
Other	9	0.84%

Table 3: Distribution of standardised language categories

These values describe the final language_group variable used in the analysis.

STANDARDISED LANGUAGE DISTRIBUTION

language_group	Count	Percentage
English	1008	94.21
Unknown	53	4.95
Other	9	0.84

Figure 6: Distribution of language categories after standardisation

.

4.4 Treatment of Missing Information

The missing-value audit conducted earlier identified missing information in *Book Name*, *language_code* and *Publishing Year*.

The missing book titles did not require imputation because *Book Name* was not intended for use as a predictive feature. The 53 missing language values were preserved through the explicit *Unknown* language category.

Only one observation contained a missing *Publishing Year*. This value was deliberately **not imputed during the general cleaning stage**. Instead, its imputation was postponed until after the training, validation and test partitions were created.

This decision ensured that the imputation value could later be learned from the training data only. Using information calculated from the full dataset before splitting could allow characteristics of the validation or test data to influence model preparation, which should be avoided when assessing generalisation performance (James et al., 2023).

This training-only preprocessing strategy will be described in detail in **Section 6**.

4.5 Construction of the Final Bestseller Target

The dataset did not contain an official binary bestseller field. As established previously, bestseller status was therefore operationally defined using units sold.

The 75th percentile remained **5,312.25 units**, and the final target was constructed as:

$$Bestseller = \begin{cases} 1, \wedge\ if unitssold < 5{,}312.25 \\ 0, \wedge\ otherwise \end{cases}$$

The final target was constructed using this threshold.

This produced the same class distribution established during the preliminary audit:

- **802 non-bestsellers (74.95%)**
- **268 bestsellers (25.05%)**

The target construction therefore retained all 1,070 observations while creating a binary classification outcome suitable for Logistic Regression.

Because units sold was used to construct the target, it could not subsequently be used as an input predictor. Other realised sales variables were also examined carefully for potential leakage before feature selection.

As illustrated previously in Figure 4, this operational definition produced a moderately imbalanced binary target containing 268 bestseller and 802 non-bestseller observations.

4.6 Summary of Cleaning Decisions

Data issue	Action taken	Reason
Artificial index variable	Removed	Sequential identifier with no predictive meaning
Repeated book titles	Investigated and retained	Records represented distinct editions or books rather than complete duplicates
genre fiction / fiction	Consolidated to fiction	Avoid artificial categorical separation
HarperCollins naming variation	Standardised	Ensure consistent publisher representation
Regional English language codes	Grouped as English	Represent the same broad language category
Missing language codes	Assigned Unknown	Preserve otherwise usable observations
Rare non-English languages	Grouped as Other	Reduce sparse categorical levels
Missing publication year	Deferred to training-only imputation	Prevent validation/test information influencing preprocessing
Bestseller target	Created using top-quartile unit sales	Produce a meaningful binary classification outcome

Table 4: Summary of data-cleaning and target-preparation decisions

4.7 Section Summary

The cleaning stage transformed the raw book dataset into a more consistent analytical dataset without unnecessarily removing legitimate observations. The artificial index variable was removed, repeated titles were investigated rather than automatically discarded, and inconsistencies in genre, publisher and language labels were standardised. Missing language information was retained using an explicit Unknown category, while the single missing publication-year value was intentionally left for later training-only imputation. Finally, bestseller status was defined using the 75th percentile of unit sales, producing a binary target suitable for Logistic Regression while ensuring that variables directly revealing the outcome could later be excluded from the predictor set.

5. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to understand the distribution of the cleaned variables, identify patterns associated with bestseller status and determine which characteristics required further investigation before model training. EDA is an important stage of predictive modelling because it allows the analyst to identify distributional characteristics, unusual observations and relationships that may influence preprocessing and model specification (Kuhn and Johnson, 2013).

The analysis focused particularly on the distribution of the binary target, the two numerical candidate predictors (*sale price* and *Publishing Year)* and the four categorical candidate predictors (*Author_Rating*, *genre*, *language_group* and *Publisher*). Visual exploration was subsequently supported by formal statistical association tests.

5.1 Bestseller Target Distribution

The first stage of the EDA examined the balance between the two outcome classes. Of the 1,070 books, **802 (74.95%)** were classified as non-bestsellers and **268 (25.05%)** as bestsellers.

This represents a class ratio of approximately **3:1**, showing that the target was moderately imbalanced. Although both classes contained enough observations for modelling, the imbalance meant that overall accuracy could not be interpreted independently. A classifier that simply predicted the majority non-bestseller class would already achieve approximately 75% accuracy without identifying any bestsellers. Metrics that account for minority-class performance, particularly precision, recall, F1-score and balanced accuracy, were therefore necessary during model evaluation (James et al., 2023).

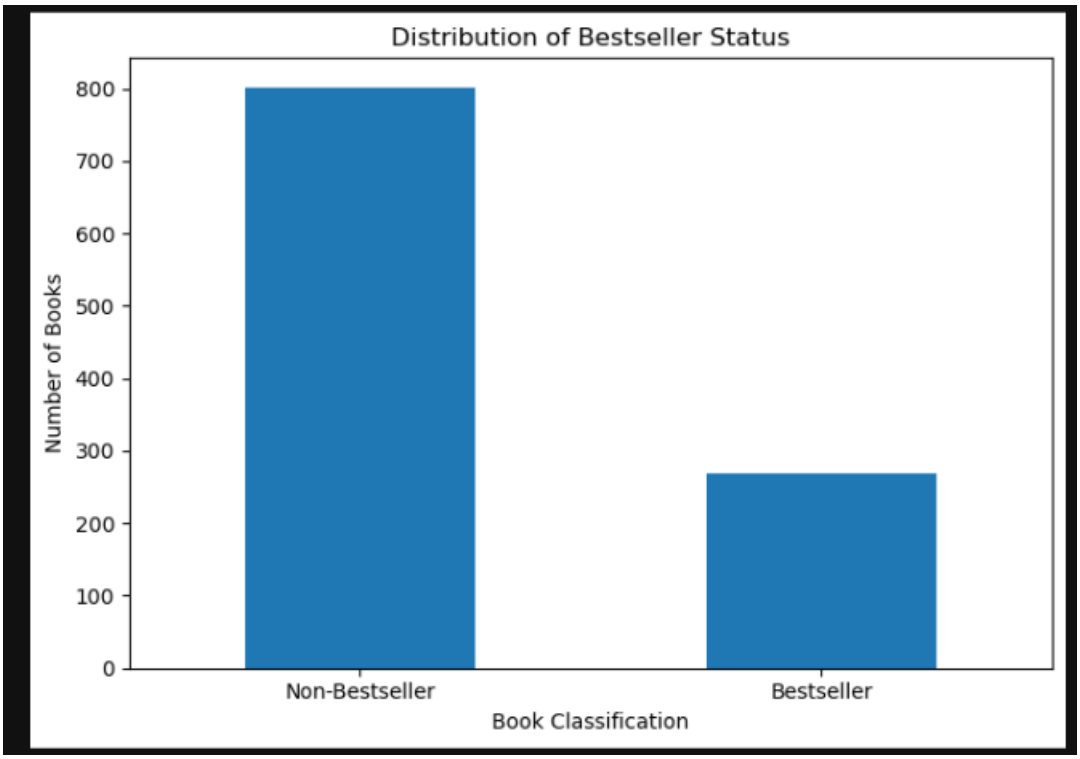


Figure 7: Distribution of bestseller and non-bestseller observations in the cleaned dataset

5.2 Numerical Predictor Distributions

5.2.1 Sale Price

The distribution of *sale price* showed that most books were concentrated at relatively low prices, with fewer observations occurring at the higher end of the range. The mean sale price was approximately **4.87**, while the median was **3.99**. Prices ranged from **0.99 to 33.86**.

The difference between the mean and median, together with the extended upper range, indicated a positively skewed distribution rather than a perfectly symmetrical one. The later decision to standardise sale price was motivated by feature-scale consistency for gradient-based optimisation; standardisation itself does not remove skewness.

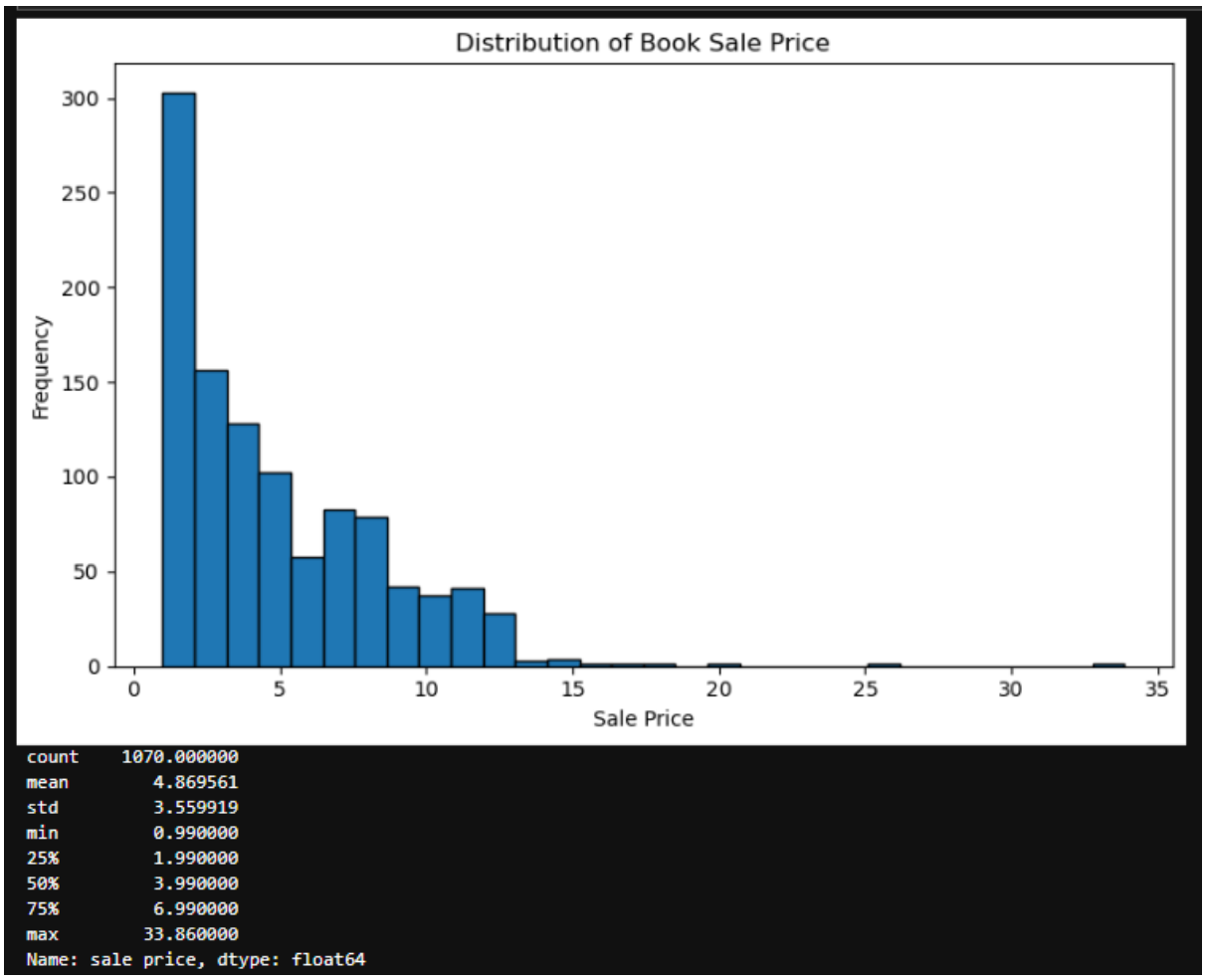


Figure 8: Distribution of book sale prices

5.2.2 Publishing Year

Publishing Year showed a particularly wide historical range. Most books were concentrated in more recent periods, although legitimate historic and ancient works created a long lower tail. As established during the dataset audit, these observations were retained because inspection indicated that they represented genuine publication periods rather than automatically being data-entry errors.

The wide scale of this variable provided further justification for standardisation before gradient-descent optimisation. Numerical variables measured on substantially different scales can make optimisation less efficient if they are supplied to a model without appropriate preprocessing (Kuhn and Johnson, 2013).

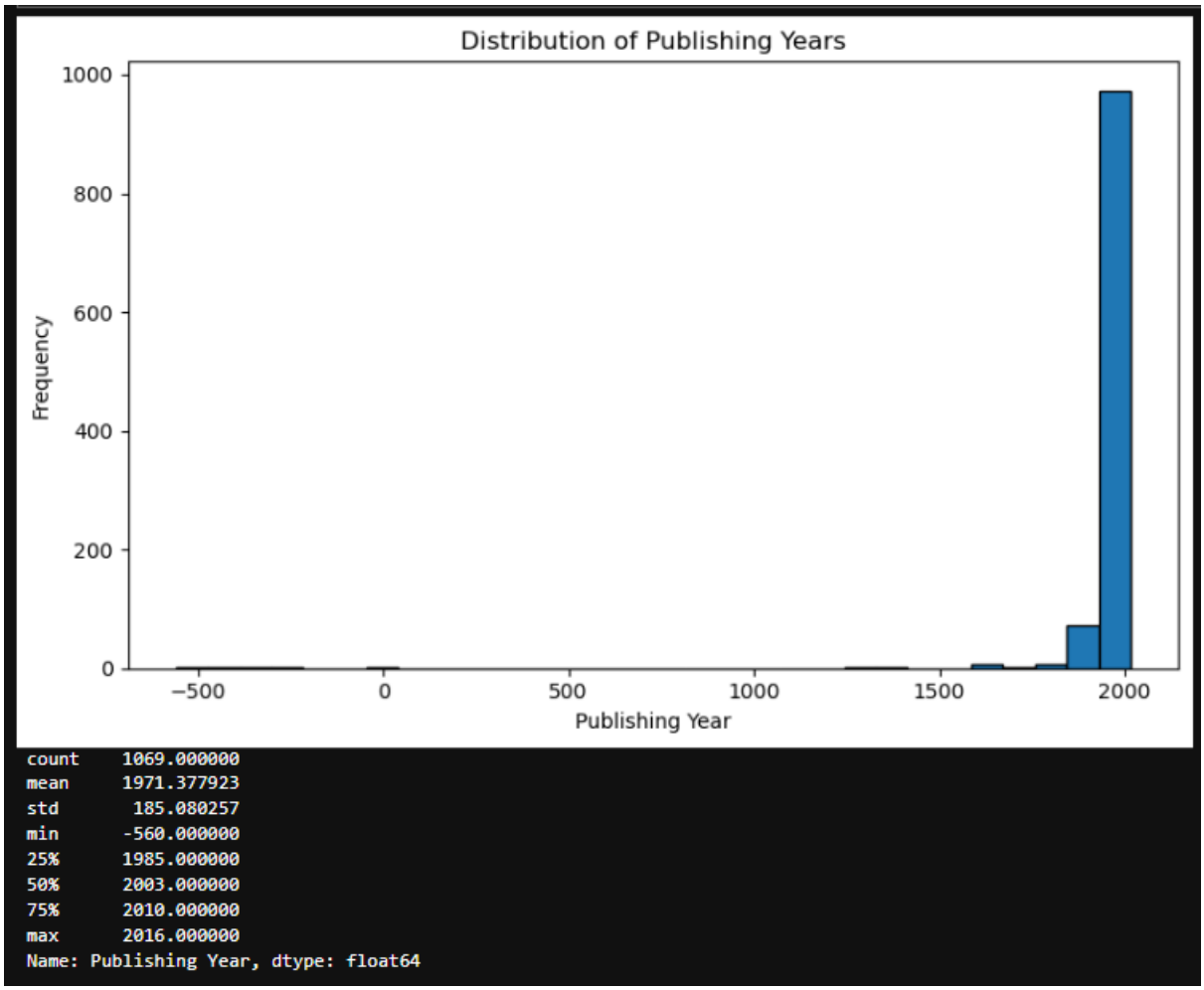


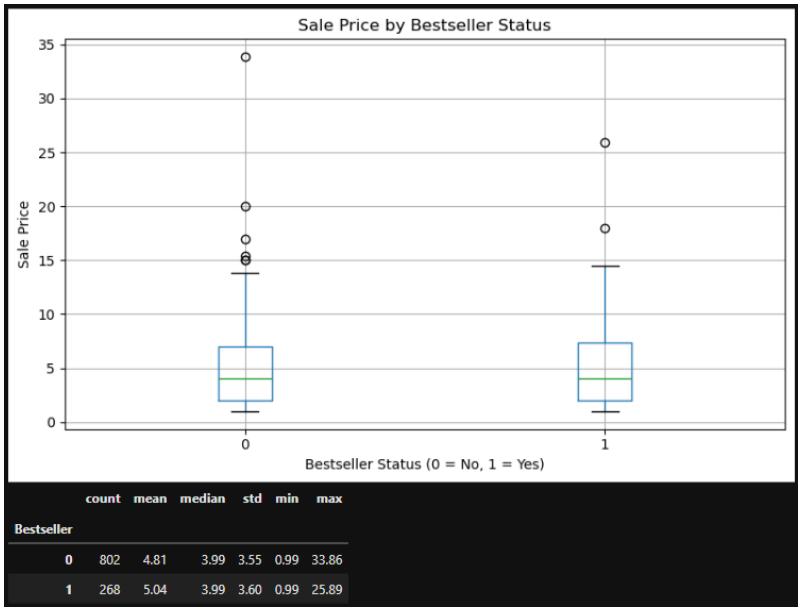
Figure 9: Distribution of book publishing years

5.3 Numerical Predictors and Bestseller Status

5.3.1 Sale Price by Bestseller Status

Non-bestsellers had an average sale price of **4.81**, compared with **5.04** among bestsellers. However, both classes had exactly the same median price of **3.99**. The distributions also overlapped substantially.

This indicated that sale price alone was unlikely to provide strong separation between bestsellers and non-bestsellers. Nevertheless, it was not immediately removed because weak univariate separation does not necessarily mean that a predictor has no value when considered jointly with other variables.



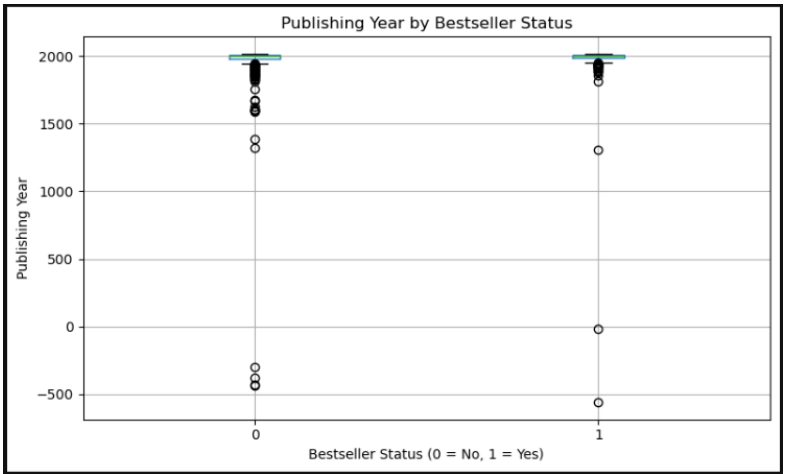
Bestseller	count	mean	median	std	min	max
0	802	4.81	3.99	3.55	0.99	33.86
1	268	5.04	3.99	3.60	0.99	25.89

Figure 10: Comparison of sale price distributions by bestseller status

5.3.2 Publishing Year by Bestseller Status

Publishing year showed a similarly high degree of overlap between the two classes. Non-bestsellers had a mean publication year of approximately **1970.91** and a median of **2002**, while bestsellers had a mean of **1972.77** and a median of **2003**.

The closeness of these central values suggested that publication year was unlikely to be a strong independent discriminator of bestseller status. However, because it represented potentially meaningful temporal information and could interact with other characteristics, it remained available for multivariable model assessment.



Bestseller	count	mean	median	std	min	max
0	801	1970.91	2002.0	178.32	-441.0	2016.0
1	268	1972.77	2003.0	204.30	-560.0	2016.0

Figure 11: Comparison of publishing year distributions by bestseller status

5.4 Categorical Predictors and Bestseller Status

Categorical predictors were investigated by calculating the proportion of books classified as bestsellers within each category. This allowed differences to be interpreted relative to the size of each group rather than simply comparing the raw number of bestsellers.

5.4.1 Author Rating

Author_Rating showed the most noticeable differences among the categorical variables. Books associated with authors rated *Excellent* had a bestseller rate of **34.81%**, while *Novice* authors had a rate of **30.00%**. The rate decreased to **20.00%** for *Intermediate* authors and **15.09%** for *Famous* authors.

These differences suggested that author rating could contain useful classification information. However, the category sizes differed considerably: the *Intermediate* group contained 625 books, whereas the *Novice* group contained only 30. Rates derived from smaller groups were therefore interpreted more cautiously

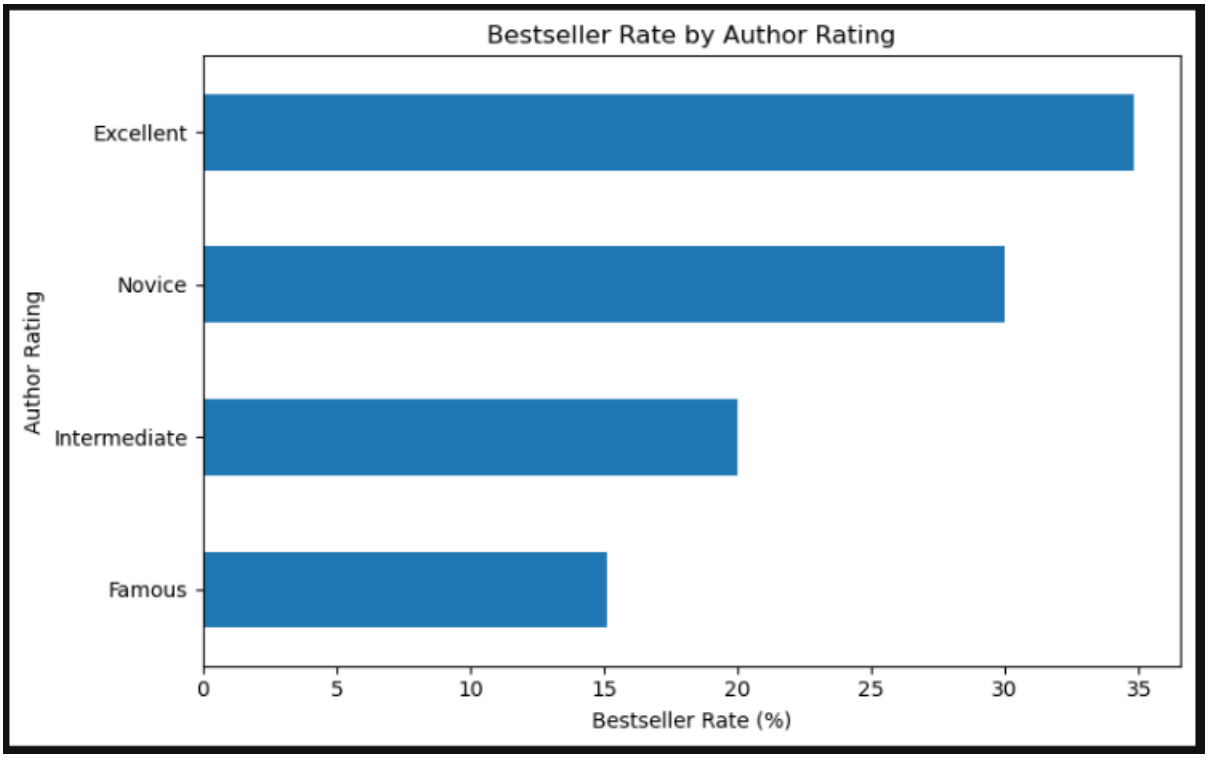


Figure 12: Bestseller rate across author-rating categories

5.4.2 Genre

Bestseller rates also varied by genre. Children's books had the highest observed rate at **46.67%**, followed by nonfiction at **26.90%** and fiction at **24.32%**. However, only **15 children's books** were present in the dataset, compared with 171 nonfiction and 884 fiction observations.

The relatively high rate for children's books should therefore not be interpreted as strong evidence that children's books are inherently more likely to become bestsellers. The small group size means that its percentage is more sensitive to individual observations.

5.4.3 Language Group

The English category contained the overwhelming majority of observations and had a bestseller rate of **25.50%**. Books with an Unknown language classification had a rate of **18.87%**, while the Other category had a rate of **11.11%**. However, the Other group contained only nine books, of which one was classified as a bestseller.

The differences therefore provided descriptive information but needed to be interpreted cautiously because of the unequal category sizes.

5.4.4 Publisher

Publisher-level bestseller rates showed moderate variation. Random House LLC had the highest observed rate among the main categories at **29.17%**, followed by Amazon Digital Services, Inc. at **26.00%**. At the lower end, Penguin Group (USA) LLC had a rate of **20.37%**, while Macmillan had **19.51%**. The very small HarperCollins Christian Publishing category contained only four observations, so its 25.00% rate should not be interpreted with the same confidence as categories containing substantially more books.

The main categorical EDA results can be summarised as follows:

Predictor	Category	Books	Bestsellers	Bestseller rate
Author Rating	Excellent	362	126	34.81%
Author Rating	Novice	30	9	30.00%
Author Rating	Intermediate	625	125	20.00%
Author Rating	Famous	53	8	15.09%
Genre	Children	15	7	46.67%
Genre	Nonfiction	171	46	26.90%
Genre	Fiction	884	215	24.32%
Language	English	1,008	257	25.50%
Language	Unknown	53	10	18.87%
Language	Other	9	1	11.11%

Table 5: Key categorical bestseller rates identified during exploratory analysis

5.5 Statistical Association Screening

Visual patterns alone were not used to make final feature-selection decisions. The EDA was therefore extended using statistical association tests.

For categorical variables, Pearson's chi-square test was used to investigate whether bestseller status was independent of the categorical predictor. Cramér's V was also calculated to describe the magnitude of the association. Numerical predictors were assessed using point-biserial correlation and the Mann–Whitney U test. The use of both visual and statistical evidence provides a stronger basis for feature screening than relying on a single descriptive measure (Kuhn and Johnson, 2013).

5.5.1 Categorical Association Tests

The categorical tests produced the following results:

Predictor	Chi-square	df	p-value	Cramér's V
Author_Rating	30.0355	3	<0.001	0.1675
genre	4.2956	2	0.1167	0.0634
language_group	2.1172	2	0.3469	0.0445
Publisher	3.8632	7	0.7954	0.0601

Table 6: Chi-square and Cramér's V results for categorical candidate predictors

The results showed that Author_Rating had the clearest statistical association with bestseller status, with \(\chi^2=30.0355\) and \(p<0.001\). Its Cramér's V of **0.1675** nevertheless indicated that the relationship was not especially strong. Genre, language group and publisher did not show statistically significant associations at the conventional 5% significance level.

5.5.2 Numerical Association Tests

The numerical tests also indicated weak univariate relationships with bestseller status.

Predictor	Point-biserial \(r\)	Correlation p-value	Mann–Whitney U	Mann–Whitney p-value
Publishing Year	0.0044	0.8868	104,303.0	0.4883
Sale price	0.0274	0.3703	103,220.5	0.3289

Table 7: Statistical association results for numerical candidate predictors

Publishing year had a point-biserial correlation of only **0.0044**, while sale price had a correlation of **0.0274**. Neither correlation was statistically significant. Similarly, neither Mann–Whitney U test produced a statistically significant result.

These results supported the earlier visual observation that sale price and publishing year showed substantial overlap between bestseller and non-bestseller books.

5.6 Chi-Square Assumption Check

The expected-frequency condition was examined before interpreting the categorical chi-square results. *Author_Rating* had **0 of 8** expected cells below five, whereas genre had **1 of 6**, language group had **1 of 6**, and publisher had **2 of 16**.

The *Author_Rating* chi-square result could therefore be interpreted with greater confidence regarding this assumption. The genre, language and publisher results were treated more cautiously because some expected counts were small. This was particularly relevant to categories such as children's books, *Other* languages and smaller publishers.

5.7 EDA Findings and Their Influence on Modelling

The EDA was used to guide subsequent modelling decisions rather than being treated as an isolated descriptive exercise.

EDA finding	Evidence	Modelling implication
Moderate target imbalance	74.95% non-bestseller vs 25.05% bestseller	Do not rely on accuracy alone; evaluate class-sensitive metrics

EDA finding	Evidence	Modelling implication
Sale price shows substantial class overlap	Both classes have median price of 3.99	Retain initially but assess contribution using validation evidence
Publishing year shows weak class separation	Medians of 2002 and 2003	Retain initially and standardise because of its wide scale
Author rating shows the clearest categorical relationship	Chi-square \(p<0.001\), Cramér's V = 0.1675	Strongest candidate for statistical feature screening
Genre differences partly reflect small groups	Children's category contains only 15 books	Interpret observed rate cautiously
Language groups are highly unequal	English = 1,008; Other = 9	Avoid overinterpreting minority categories
Publisher rates vary moderately	Approximately 19.51%–29.17% among major groups	Validate usefulness in multivariable model
Several predictors are weak individually	Non-significant univariate results	Do not automatically discard them before multivariable validation

Table 8: Key exploratory findings and their influence on subsequent model development

Importantly, the absence of a statistically significant univariate relationship was **not treated as automatic evidence that a predictor should be removed**. Logistic Regression estimates effects while predictors are considered jointly, and a variable with limited standalone separation may still contribute useful information in combination with others (James et al., 2023). The subsequent feature-selection stage therefore compared a fuller leakage-safe predictor set with a statistically reduced alternative using validation performance.

5.8 Section Summary

The exploratory analysis showed that the bestseller target was moderately imbalanced, with approximately one bestseller for every three non-bestsellers. Sale price and publishing year displayed substantial overlap between the classes and produced weak univariate statistical associations. In contrast, *Author_Rating* showed the clearest relationship with bestseller status, although its effect size remained modest. Genre, language and publisher displayed descriptive differences, but several small categories required cautious interpretation. These findings indicated that no single variable was likely to provide strong classification performance independently. Consequently, the subsequent modelling strategy retained a broader leakage-safe predictor set initially and used validation-based comparison to determine whether a reduced feature set improved performance.

6. FEATURE SELECTION AND MODELLING PREPARATION

Following the exploratory analysis, the next stage prepared the data for Logistic Regression while controlling for target leakage, post-outcome information and information leakage between the training, validation and test partitions. Feature selection was not based solely on statistical significance. Instead, variables were first assessed according to whether they would realistically be available when estimating a book's bestseller potential, after which statistical evidence and later validation performance were used to assess their usefulness.

This approach was important because a predictor can show a weak relationship with the target when examined independently but may still contribute useful information when combined with other variables in a multivariable Logistic Regression model (James et al., 2023).

6.1 Feature-Selection Strategy

The original dataset contained several variables that would not be appropriate for an early bestseller-prediction model. Three main exclusion categories were identified.

Book Name and *Author* were excluded as identifiers and high-cardinality fields. The dataset contained 1,046 unique book names and 735 unique authors, meaning that direct one-hot encoding of these variables would create a large and highly sparse predictor space relative to the 1,070 available observations. More importantly, they identify individual books or authors rather than providing generalisable characteristics.

units sold, gross sales, publisher revenue and sales rank were excluded because they directly describe realised commercial performance. Since bestseller status was derived from units sold, using these variables would allow the predictor matrix to contain information closely related to the outcome it was attempting to predict. They were therefore classified as sales-leakage variables.

Book_average_rating and *Book_ratings_count* were also excluded from the initial model because they represent reader-response information generated after readers have interacted with a book. Such variables may not be available at the point where a publisher wishes to make an early commercial prediction.

The resulting variable-selection decisions are summarised below.

Variable/group	**Decision**	**Reason**
Publishing Year	Retain	Potentially relevant temporal characteristic
sale price	Retain	Commercial feature potentially available before realised sales
Author_Rating	Retain	Showed strongest univariate categorical association
genre	Retain	Potentially relevant market characteristic
Publisher	Retain	Publisher may influence market reach and positioning
language_group	Retain	General book characteristic available before outcome
Book Name	Exclude	Identifier/high cardinality

Variable/group	Decision	Reason
Author	Exclude	Identifier/high cardinality
units sold	Exclude	Used to define target; direct leakage
gross sales	Exclude	Realised sales information
publisher revenue	Exclude	Realised commercial outcome
sales rank	Exclude	Direct measure of realised market performance
Book_average_rating	Exclude	Post-publication reader-response information
Book_ratings_count	Exclude	Post-publication reader-response information

Table 9: Feature-selection decisions before model training

The final raw predictor matrix therefore contained **six predictors**: Publishing Year, sale price, Author_Rating, genre, Publisher and language_group, with Bestseller as the target. The modelling matrix initially contained 1,070 observations and six raw predictor columns.

Although the statistical screening in Section 5 found non-significant univariate results for several of these variables, they were not automatically removed. The broader leakage-safe predictor set was retained so that its usefulness could later be compared with a reduced feature set using validation performance.

This is preferable to performing mechanical elimination using p-values alone because statistical significance and predictive usefulness are not necessarily identical objectives (Kuhn and Johnson, 2013).

6.2 Training, Validation and Test Partitioning

The 1,070 observations were divided into separate training, validation and test partitions using a **70:15:15** strategy. A fixed *random_state* of 42 was used to make the split reproducible.

The resulting partition sizes were:

Partition	Observations	Approximate proportion
Training	749	70%
Validation	160	15%
Test	161	15%
Total	**1,070**	**100%**

Table 10: Final training, validation and test partition sizes

The resulting partition sizes were 749 training, 160 validation and 161 test observations.

The purpose of each partition was kept distinct. The **training set** was used to estimate model parameters and learn preprocessing parameters. The **validation set** was used for

feature-selection, threshold and model-development decisions. The **test set** was kept untouched until the final model configuration had been selected. This separation reduces the risk that repeated model-development decisions become indirectly tailored to the final test observations (Géron, 2022).

Because the target distribution was moderately imbalanced, stratified sampling was applied. Stratification preserved approximately the same bestseller proportion across all three datasets:

Partition	Non-bestseller	Bestseller	Bestseller proportion
Training	561	188	25.10%
Validation	120	40	25.00%
Test	121	40	24.84%

Table 11: Target-class representation across the modelling partitions

The near-identical class proportions confirm that the stratification procedure worked as intended.

This was particularly important because a non-stratified random split could have produced different bestseller proportions across the three relatively small partitions, making performance comparisons less reliable.

	Partition	Class	Count	Percentage
0	Training	Non-Bestseller	561	74.90
1	Training	Bestseller	188	25.10
2	Validation	Non-Bestseller	120	75.00
3	Validation	Bestseller	40	25.00
4	Test	Non-Bestseller	121	75.16
5	Test	Bestseller	40	24.84

Figure 13: Target distribution across Training, Validation and Test partitions

6.3 Numerical and Categorical Feature Groups

The six selected predictors required different preprocessing procedures and were therefore divided into numerical and categorical groups.

The numerical features were:

- *Publishing Year*

- *sale price*

The categorical features were:

- *Author_Rating*
- *genre*
- *Publisher*
- *language_group*

These were the feature groups used in the preprocessing pipeline.

Separating the features in this manner allowed a ColumnTransformer to apply appropriate preprocessing to each data type without altering the original semantic meaning of the variables.

6.4 Training-Only Preprocessing

A major methodological consideration was ensuring that all preprocessing parameters were learned **only from the training partition**. The validation and test data were transformed using parameters learned from the training set rather than being used to estimate their own transformation values.

This procedure reduces preprocessing leakage. If information from the validation or test sets were used to calculate medians, means, standard deviations or categorical encodings, the model-development process would indirectly have access to information that should remain unseen (Géron, 2022).

6.4.1 Missing-Value Imputation

Only one selected numerical observation had a missing value: one *Publishing Year*. Rather than removing an otherwise usable observation, a *SimpleImputer* using the **median** strategy was included in the numerical pipeline.

Importantly, the median was fitted using the training data only. The same training-derived value was then used wherever necessary when transforming later partitions.

Median imputation was appropriate because the publishing-year distribution contained a wide historical range and extreme old values. Compared with the mean, the median is less influenced by unusually distant observations.

6.4.2 Numerical Standardisation

Following imputation, *Publishing Year* and *sale price* were standardised using *StandardScaler*.

For each numerical variable, standardisation transforms observations according to:

$$z = \frac{x - \mu}{\sigma}$$

where:

- x represents the original observation;
- μ represents the training-set mean; and
- σ represents the training-set standard deviation.

This results in numerical features that are approximately centred around zero with unit variance.

Standardisation was especially useful for this project because the Logistic Regression model was trained using manually implemented gradient descent. *Publishing Year* and *sale price* are measured on very different numerical scales, and scaling prevents one feature's numerical magnitude from dominating gradient calculations simply because of its unit of measurement (Kuhn and Johnson, 2013).

6.5 One-Hot Encoding of Categorical Variables

The four categorical predictors could not be supplied directly to the numerical Logistic Regression algorithm. They were therefore transformed using one-hot encoding.

The categorical encoder was configured with:

- *drop="first"*
- *handle_unknown="ignore"*
- *sparse_output=False*

within the categorical preprocessing pipeline.

drop="first" meant that one category from each variable was omitted and treated as the reference level. This prevented a complete set of redundant dummy variables from being included in the Logistic Regression design matrix.

The reference categories produced by the fitted encoder were:

Categorical predictor	Reference category
Author_Rating	Excellent
genre	children
Publisher	Amazon Digital Services, Inc.
language_group	English

Table 12: Reference categories used in one-hot encoding

These reference categories were verified from the fitted preprocessing pipeline.

As a result, later Logistic Regression coefficients for the remaining dummy variables were interpreted relative to these omitted reference categories.

6.6 Final Processed Feature Space

The preprocessing pipeline was fitted on the 749 training observations and then applied unchanged to the validation and test partitions.

After preprocessing, the matrices had the following dimensions:

Partition	Rows	Processed features
Training	749	16
Validation	160	16
Test	161	16

Table 13: Dimensions of the processed modelling matrices

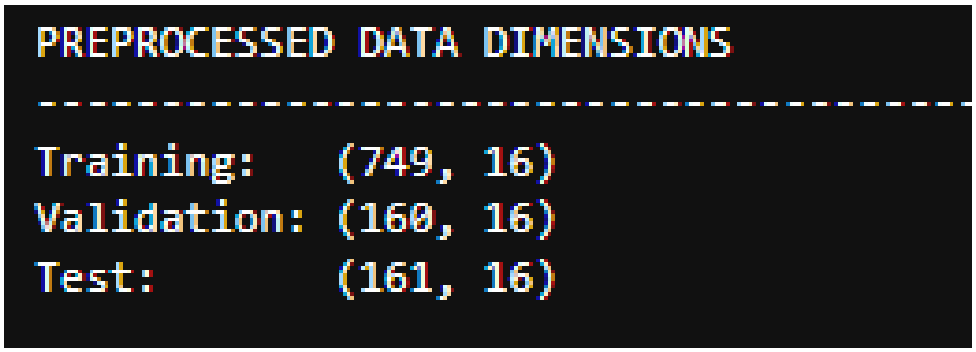


Figure 14: Preprocessed Data Dimensions

All three partitions contained the same 16 processed predictor columns.

The complete processed feature set was:

No.	Processed feature
1	Publishing Year
2	sale price
3	Author_Rating_Famous
4	Author_Rating_Intermediate
5	Author_Rating_Novice
6	genre_fiction
7	genre_nonfiction
8	Publisher_Hachette Book Group
9	Publisher_HarperCollins Christian Publishing
10	Publisher_HarperCollins Publishers
11	Publisher_Macmillan
12	Publisher_Penguin Group (USA) LLC
13	Publisher_Random House LLC

No.	Processed feature
14	Publisher_Simon and Schuster Digital Sales Inc
15	language_group_Other
16	language_group_Unknown

Table 14: Final processed predictors supplied to the Logistic Regression algorithm

The exact 16 processed variables were retrieved directly from the fitted preprocessing pipeline.

The two original numerical predictors remained as continuous standardised features, while the four categorical predictors expanded into 14 dummy variables after reference categories were removed.

```
PROCESSED MODEL FEATURES
------------------------------------------------------------
 1. Publishing Year
 2. sale price
 3. Author_Rating_Famous
 4. Author_Rating_Intermediate
 5. Author_Rating_Novice
 6. genre_fiction
 7. genre_nonfiction
 8. Publisher_Hachette Book Group
 9. Publisher_HarperCollins Christian Publishing
10. Publisher_HarperCollins Publishers
11. Publisher_Macmillan
12. Publisher_Penguin Group (USA) LLC
13. Publisher_Random House LLC
14. Publisher_Simon and Schuster Digital Sales Inc
15. language_group_Other
16. language_group_Unknown

Total processed features: 16
```

Figure 15: Processed Feature List

6.7 Final Preprocessing Verification

Before model development began, the resulting numerical matrices were checked for invalid values.

The verification confirmed:

- 0 missing values in the training matrix;
- 0 missing values in the validation matrix;
- 0 missing values in the test matrix;
- 0 infinite values in the training matrix;
- 0 infinite values in the validation matrix; and
- 0 infinite values in the test matrix.

This ensured that all three partitions could be supplied directly to the manually developed Logistic Regression functions without additional cleaning.

It is also important to clarify that although scikit-learn utilities were used for **data partitioning, imputation, scaling and one-hot encoding**, the Logistic Regression classifier itself was **not** implemented using *sklearn.linear_model.LogisticRegression*. The classification algorithm was implemented manually using NumPy, including calculation of the linear predictor, sigmoid transformation, binary cross-entropy loss, gradients and iterative gradient-descent parameter updates. The preprocessing tools therefore supported data preparation rather than replacing the required from-scratch model.

6.8 Feature-Selection Position Before Model Training

At this stage, the model-development dataset contained the full **16-feature leakage-safe processed representation**.

Although *Author_Rating* had provided the strongest statistical evidence during univariate screening, the other candidate variables were deliberately retained for the first model. This avoided making feature-selection decisions entirely on the basis of univariate p-values.

The approach followed in the remainder of the analysis was therefore:

1. train the baseline Logistic Regression using the full processed predictor set;
2. evaluate performance on validation data;
3. investigate class imbalance, classification threshold and optimisation behaviour;
4. construct a reduced feature alternative based on the earlier statistical screening; and
5. compare the full and reduced alternatives using the same predeclared validation criterion before selecting the final model.

This approach integrates both statistical feature screening and predictive validation. It also avoids using the final test set to decide which predictors should remain in the model.

6.9 Section Summary

The feature-selection stage reduced the original dataset to six leakage-safe raw predictors while excluding identifiers, realised sales measures and post-publication reader-response variables. The 1,070 observations were stratified into 749 training, 160 validation and 161 test records, preserving an approximately 25% bestseller rate in each partition. Numerical variables were median-imputed where required and standardised, while categorical variables were one-hot encoded with one reference category removed from each field. All preprocessing parameters were fitted using training data only. The six raw predictors consequently produced **16 numerical model features**, and final validation confirmed that the resulting training, validation and test matrices contained no missing or infinite values.

The prepared matrices were therefore ready for the central requirement of the assessment: **developing Logistic Regression from scratch**.

7. LOGISTIC REGRESSION MODEL DEVELOPMENT FROM SCRATCH

The central technical requirement of the analysis was to develop a Logistic Regression classifier **from scratch** rather than relying on a pre-built Logistic Regression estimator. The model was therefore implemented manually using NumPy. Scikit-learn utilities were used only during the preceding stages for partitioning and preprocessing; the classification algorithm itself manually implemented the linear predictor, sigmoid function, binary cross-entropy loss, gradient calculation, batch gradient descent and prediction functions.

Logistic Regression is appropriate for binary classification because it models the probability that an observation belongs to one of two outcome classes. Unlike ordinary linear regression, which can produce unrestricted numerical outputs, Logistic Regression applies the logistic or sigmoid transformation so that predictions fall between zero and one (James et al., 2023).

7.1 Preparation of NumPy Matrices

The manually developed algorithm was designed to operate directly on numerical NumPy arrays. The previously transformed predictor matrices and target vectors were therefore converted into floating-point arrays before training.

The resulting dimensions were:

Dataset	Predictor dimensions	Target dimensions
Training	(749, 16)	(749,)
Validation	(160, 16)	(160,)
Test	(161, 16)	(161,)

Table 15: Dimensions of the NumPy arrays used by the from-scratch Logistic Regression algorithm

These dimensions confirm that each of the 749 training books was represented by the same 16 processed predictor variables identified in Section 6.

7.2 Linear Predictor and Sigmoid Function

For each observation, Logistic Regression first calculates a linear combination of the input features:

$$[z = X\omega + b]$$

where:

- X represents the predictor matrix;
- ω represents the vector of model weights; and
- b represents the intercept.

The resulting value z is unrestricted and can range from negative to positive infinity. It is therefore converted into a probability using the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The output of the sigmoid function lies between 0 and 1 and can therefore be interpreted as the estimated probability that a book belongs to the bestseller class (James et al., 2023).

The sigmoid function was manually implemented as:

$$sigmoid(z) = 1/(1 + exp(-z))$$

For numerical stability, the implementation clipped extreme linear values to the interval -500 to 500 before applying the exponential calculation. This prevents very large positive or negative inputs from producing numerical overflow.

The function was then tested using a simple range of values. For example, a linear value of 0 produced a probability of exactly 0.5000, while increasingly positive values approached 1 and increasingly negative values approached 0.

	Linear Value (z)	Sigmoid Probability
0	-5.0	0.0067
1	-1.0	0.2689
2	0.0	0.5000
3	1.0	0.7311
4	5.0	0.9933

Figure 16: Verification of the manually implemented sigmoid probability transformation

7.3 Binary Cross-Entropy Loss

After predicted probabilities were generated, the difference between the predictions and the observed target classes had to be quantified. Binary cross-entropy was implemented as the optimisation loss function.

For m observations:

$$J(w, b) = \frac{-1}{m} \sum_{i=1}^{m} [y_i log(p_i) + (1 - y_i) log(1 - p_i)]$$

where:

- y_i represents the observed class;
- p_i represents the predicted probability; and
- m represents the number of observations.

Binary cross-entropy penalises predictions that assign low probabilities to the true class. The objective of training is therefore to find parameter values that minimise this loss (James et al., 2023).

The function was manually implemented using NumPy. To avoid attempting to calculate $log(0)$, predicted probabilities were clipped using a small constant of:

$$\epsilon = 10^{-15}$$

The implementation therefore included its own numerical protection rather than relying on an external optimisation function.

The loss function was tested using deliberately strong and weak predictions. Strong predictions produced a loss of **0.1339**, while weaker predictions produced a substantially larger loss of **0.6259**. This confirmed that the function correctly penalised poorer probability estimates.

7.4 Gradient Calculation

Gradient descent requires calculating how the loss changes when each model parameter changes. These gradients were also implemented manually.

The gradient for the weight vector was calculated as:

$$\frac{\partial J}{\partial w} = \frac{1}{m} X^T (p - y)$$

while the intercept gradient was calculated as:

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum (p - y)$$

where $p - y$ represents the prediction error across the observations.

The NumPy implementation first calculated:

$$errors = probabilities - y$$

and then obtained the gradients through matrix multiplication and averaging.

This stage is especially important for demonstrating that the model was developed from scratch. The implementation did not call a Logistic Regression fitting routine to estimate the coefficients; instead, the required derivatives were explicitly calculated and supplied to the optimisation procedure.

7.5 Batch Gradient-Descent Training Algorithm

The separate mathematical components were combined into a complete training function.

The model weights were initially set to zero:

$$w_0 = 0$$

and the intercept was also initialised to:

$$b_0 = 0$$

Each training iteration then followed five stages:

1. calculate the linear predictor $z = Xw + b$;
2. transform the linear values into probabilities using the sigmoid function;
3. calculate binary cross-entropy loss;
4. calculate the weight and intercept gradients; and
5. update the model parameters.

The gradient-descent updates were:

$$w := w - \alpha \frac{\partial J}{\partial w}$$

and:

$$b := b - \alpha \frac{\partial J}{\partial b}$$

Where α represents the learning rate.

Because the gradients were calculated using the entire training dataset before each update, the approach represented **batch gradient descent** rather than stochastic or mini-batch gradient descent.

The training function also recorded the training loss after every iteration. When validation data were supplied, validation loss was calculated for monitoring purposes, but validation observations were **not used to calculate the gradient updates**. This maintained the intended separation between parameter estimation and model-development evaluation.

The training implementation explicitly performed the forward pass, loss calculation, gradient computation, weight update, intercept update and validation-loss calculation manually.

```
# ------------------------------------------
# Forward pass
# ------------------------------------------

linear_output = (
    X_train @ weights + bias
)

train_probabilities = sigmoid(
    linear_output
)

# ------------------------------------------
# Calculate training loss
# ------------------------------------------

train_loss = binary_cross_entropy(
    y_train,
    train_probabilities
)

training_loss_history.append(
    train_loss
)

# ------------------------------------------
# Calculate gradients
# ------------------------------------------

dw, db = compute_gradients(
    X_train,
    y_train,
    train_probabilities
)

# ------------------------------------------
# Gradient-descent update
# ------------------------------------------

weights -= learning_rate * dw
bias -= learning_rate * db
```

Figure 17: Core implementation of the from-scratch batch gradient-descent training algorithm

7.6 Probability and Class Prediction

Separate prediction functions were developed once the training algorithm had been completed.

The probability function first calculated:

$$z = Xw + b$$

and passed the result through the previously implemented sigmoid function.

The class-prediction function subsequently compared each probability with a classification threshold. For the initial baseline model, the conventional threshold of **0.50** was used:

$$\acute{y} = \begin{cases} 1, & p \geq 0.50 \\ 0, & p < 0.50 \end{cases}$$

where:

- 1 represented a bestseller; and
- 0 represented a non-bestseller.

The 0.50 threshold was treated only as the **baseline decision rule**. It was not assumed to be optimal. Later validation analysis would determine whether a different threshold produced a more appropriate balance between precision and recall.

7.7 Baseline Model Configuration

Before training, the baseline hyperparameters were deliberately defined rather than repeatedly adjusted without evidence.

The initial configuration was:

Hyperparameter	Baseline setting
Learning rate	0.05
Maximum iterations	5,000
Classification threshold	0.50
Optimisation method	Batch gradient descent
Initial weights	0
Initial intercept	0

Table 16: Initial configuration of the from-scratch Logistic Regression model

The purpose of this configuration was to establish a reproducible reference model. Any later modification would have to be justified using validation evidence rather than by repeatedly altering the parameters until a desirable result was obtained.

The learning rate, number of iterations, threshold and optimisation method were defined before model fitting.

7.8 Baseline Model Training

The baseline model was trained using the **749 training observations and 16 processed predictor variables**.

Training began with a binary cross-entropy loss of:

$$0.693147$$

and ended after 5,000 iterations with a training loss of:

$$0.548574$$

The corresponding final validation loss was:

$$0.558452$$

The model learned **16 coefficients** and an intercept of:

$$-0.269940$$

These results confirm that the optimisation procedure changed the initially zero-valued parameters and substantially reduced the loss from its starting value.

Training result	Value
Initial training loss	0.693147
Final training loss	0.548574
Final validation loss	0.558452
Learned intercept	-0.269940
Learned weights	16

Table 17: Baseline Logistic Regression training results

The relatively small difference between the final training and validation loss at this stage suggested that optimisation had learned a pattern that transferred reasonably consistently from the training observations to the validation observations in terms of probability loss. However, loss convergence alone does **not** demonstrate good classification performance. Class predictions and evaluation metrics still had to be examined separately.

7.9 Training-Loss Behaviour

Training and validation loss were plotted across the 5,000 gradient-descent iterations to visually assess optimisation behaviour.

The loss curve demonstrated that the manually implemented optimisation process reduced the objective function progressively rather than exhibiting unstable oscillation or divergence. This provided additional evidence that the learning rate of 0.05 was sufficiently stable for the standardised feature space.

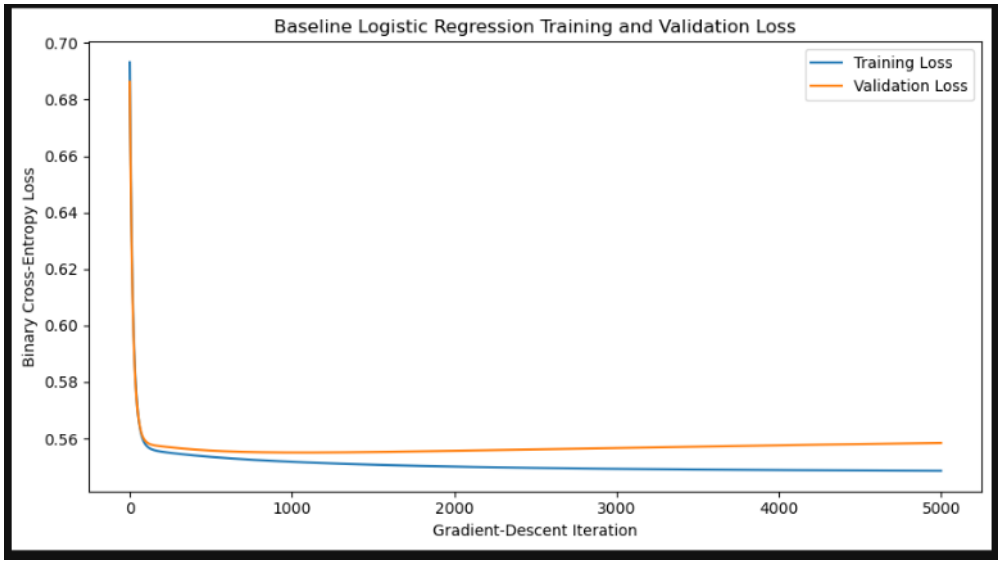


Figure 18: Training and validation binary cross-entropy loss across baseline gradient-descent iterations

At this point, however, the loss graph should not be interpreted as evidence that the classifier was already effective at identifying bestsellers. The next stage evaluates the actual validation classifications using class-sensitive performance metrics.

7.10 Verification of the From-Scratch Requirement

The completed implementation demonstrates that Logistic Regression was manually developed through the following components:

Component	Implementation
Linear predictor	$Xw + b$ calculated directly using NumPy
Probability transformation	Custom sigmoid function
Loss function	Custom binary cross-entropy
Gradient for weights	Manually calculated using matrix multiplication
Gradient for intercept	Manually calculated from prediction errors
Parameter initialisation	Zero-valued weights and intercept
Optimisation	Custom batch gradient-descent loop
Probability prediction	Custom prediction function
Class prediction	Custom probability-threshold function
Pre-built Logistic Regression estimator	Not used

Table 18: Components demonstrating the from-scratch Logistic Regression implementation

The model-development stage therefore satisfied the requirement to train Logistic Regression from scratch while still allowing standard data-processing utilities to be used before the algorithm received its numerical input.

7.11 Section Summary

A complete binary Logistic Regression classifier was developed manually using NumPy. The implementation calculated the linear predictor, converted it to probabilities using a custom sigmoid function, measured prediction error using manually implemented binary cross-

entropy and calculated the gradients required for optimisation. The weights and intercept were then learned using batch gradient descent rather than a pre-built Logistic Regression estimator. Using a learning rate of 0.05 over 5,000 iterations reduced training loss from **0.693147 to 0.548574**, while final validation loss was **0.558452**. The training-loss behaviour confirmed that the optimisation procedure was functioning as intended. The next stage therefore evaluates whether the resulting probabilities produced useful bestseller classifications.

8. BASELINE MODEL EVALUATION

After confirming that the from-scratch Logistic Regression algorithm converged successfully, the next stage evaluated its predictive performance using the **validation partition only**. The final test set remained untouched because decisions concerning threshold adjustment, class weighting and feature refinement still needed to be made.

Evaluating a classifier using several complementary metrics is particularly important when the target classes are imbalanced. Accuracy alone can provide a misleading impression of performance if a model simply predicts the majority class repeatedly (Géron, 2022). The baseline model was therefore assessed using accuracy, balanced accuracy, precision, recall, specificity, F1-score, confusion matrix and ROC-AUC.

8.1 Evaluation Metrics

The following metrics were used to evaluate the binary classifier.

Accuracy measures the proportion of all observations that are classified correctly:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where TP, TN, FP and FN represent true positives, true negatives, false positives and false negatives respectively.

Precision measures the proportion of predicted bestsellers that are actually bestsellers:

$$Precision = \frac{TP}{TP + FP}$$

A high precision score therefore indicates that the model produces relatively few false bestseller predictions.

Recall, also referred to as sensitivity, measures the proportion of actual bestsellers that the model successfully identifies:

$$Recall = \frac{TP}{TP + FN}$$

Specificity measures the proportion of actual non-bestsellers correctly identified:

$$Specificity = \frac{TN}{TN + FP}$$

The **F1-score** combines precision and recall using their harmonic mean:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Balanced accuracy gives equal importance to the ability to classify each target class and is useful when class frequencies differ substantially. ROC-AUC, in contrast, evaluates how effectively predicted probabilities rank positive observations above negative observations across multiple possible classification thresholds (James et al., 2023).

8.2 Baseline Validation Predictions

Before calculating the evaluation metrics, the predicted validation probabilities were examined.

Using the baseline classification threshold of **0.50**, the model predicted:

- **160 non-bestsellers**
- **0 bestsellers**

The predicted probabilities ranged from **0.1317 to 0.4338**, with an overall mean probability of **0.2462**.

The most important finding was that the **maximum predicted probability was only 0.4338**, which was below the default classification threshold of 0.50. Consequently, no validation observation could be assigned to the bestseller class.

This did not necessarily mean that the model had learned no information. Rather, it immediately suggested that the default threshold could be unsuitable for the probability distribution produced by the model.

8.3 Baseline Validation Performance

The baseline classifier produced the following validation results:

Metric	Validation score
Accuracy	0.7500
Balanced Accuracy	0.5000
Precision	0.0000
Recall	0.0000
Specificity	1.0000
F1-Score	0.0000
ROC-AUC	0.5724

Table 19: Baseline Logistic Regression validation performance at a 0.50 classification threshold

These metrics were obtained from the baseline validation evaluation.

At first glance, the **75.00% accuracy** could appear acceptable. However, this result was highly misleading. The model classified every observation as a non-bestseller and therefore achieved its accuracy simply because non-bestsellers represented 75% of the validation data.

The minority bestseller class was not successfully identified at all. Precision, recall and F1-score were therefore all zero. Specificity was 1.0000 because every one of the 120 actual non-bestsellers was correctly assigned to the negative class.

This demonstrates why accuracy should not be considered in isolation when evaluating an imbalanced classification problem (Géron, 2022).

8.4 Majority-Class Benchmark

To place the 75% accuracy into context, a naïve majority-class benchmark was created.

The benchmark made no attempt to identify bestseller patterns. It simply predicted every validation observation as a non-bestseller.

Its performance was:

Metric	Majority-class benchmark	Baseline Logistic Regression
Accuracy	0.7500	0.7500
Balanced Accuracy	0.5000	0.5000
Precision	0.0000	0.0000
Recall	0.0000	0.0000
F1-Score	0.0000	0.0000

Table 20: Comparison of baseline Logistic Regression with the majority-class benchmark

The majority benchmark and baseline model produced identical threshold-based classification performance.

This comparison is important because it confirms that the baseline model's 75% accuracy did **not** represent useful bestseller classification. A model that required no predictor information whatsoever could achieve the same accuracy simply by exploiting the class distribution.

For this reason, raw accuracy was not selected as the primary criterion for subsequent model improvement.

8.5 Baseline Confusion Matrix

The baseline confusion matrix provided a clearer representation of the classification problem.

The model produced:

- **True Negatives (TN): 120**
- **False Positives (FP): 0**
- **False Negatives (FN): 40**
- **True Positives (TP): 0**

Therefore, all **120 non-bestsellers** were classified correctly, but all **40 actual bestsellers** were missed.

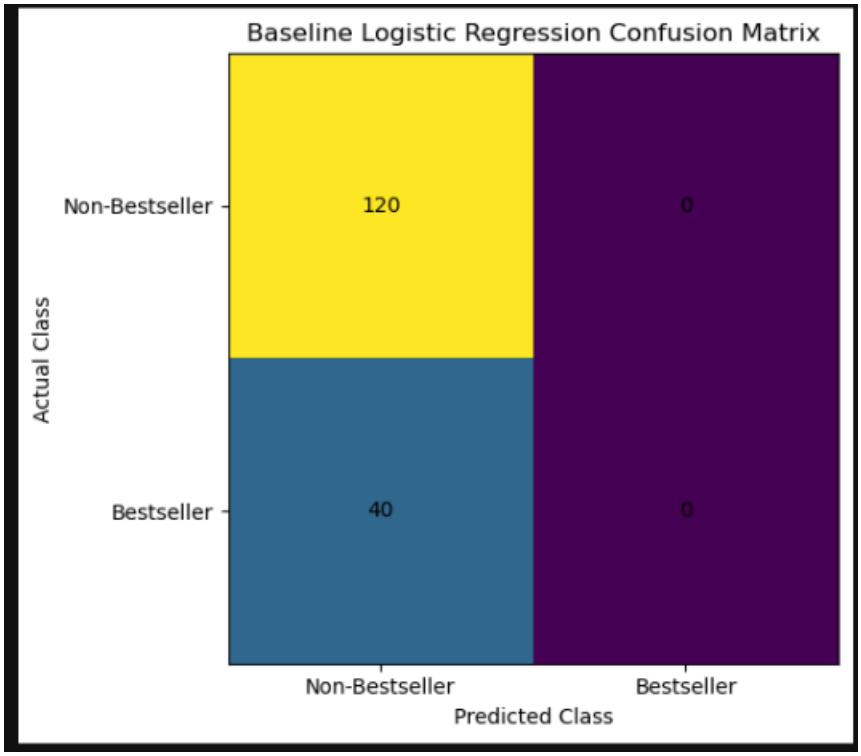


Figure 19: Baseline Logistic Regression validation confusion matrix at a 0.50 threshold

The confusion matrix demonstrates the limitation of the baseline classifier more directly than accuracy. The model had perfect specificity because it produced no false positives, but this occurred at the cost of **zero recall** for the bestseller class.

In a publishing context, this behaviour would be impractical if the objective were to identify books with potential to achieve strong sales. A screening model that fails to identify every actual bestseller would provide little decision-making value despite reporting 75% accuracy.

8.6 ROC-AUC Evaluation

Threshold-dependent metrics showed that the baseline classification rule was ineffective. However, it was also important to determine whether the predicted **probabilities** contained any useful ranking information before modifying the model.

The Receiver Operating Characteristic (ROC) curve evaluates the trade-off between the true-positive rate and false-positive rate across different classification thresholds. The corresponding area under the curve, ROC-AUC, provides a threshold-independent measure of discrimination (James et al., 2023).

The baseline model achieved:

$$ROC - AUC = 0.5724$$

A value of 0.50 corresponds approximately to random ranking, while larger values indicate greater ability to separate the positive and negative classes. The baseline score of **0.5724** was therefore only moderately above random ranking.

The result nevertheless showed that the model's probabilities contained **some limited discriminatory information**, even though the default threshold converted all of them into non-bestseller classifications.

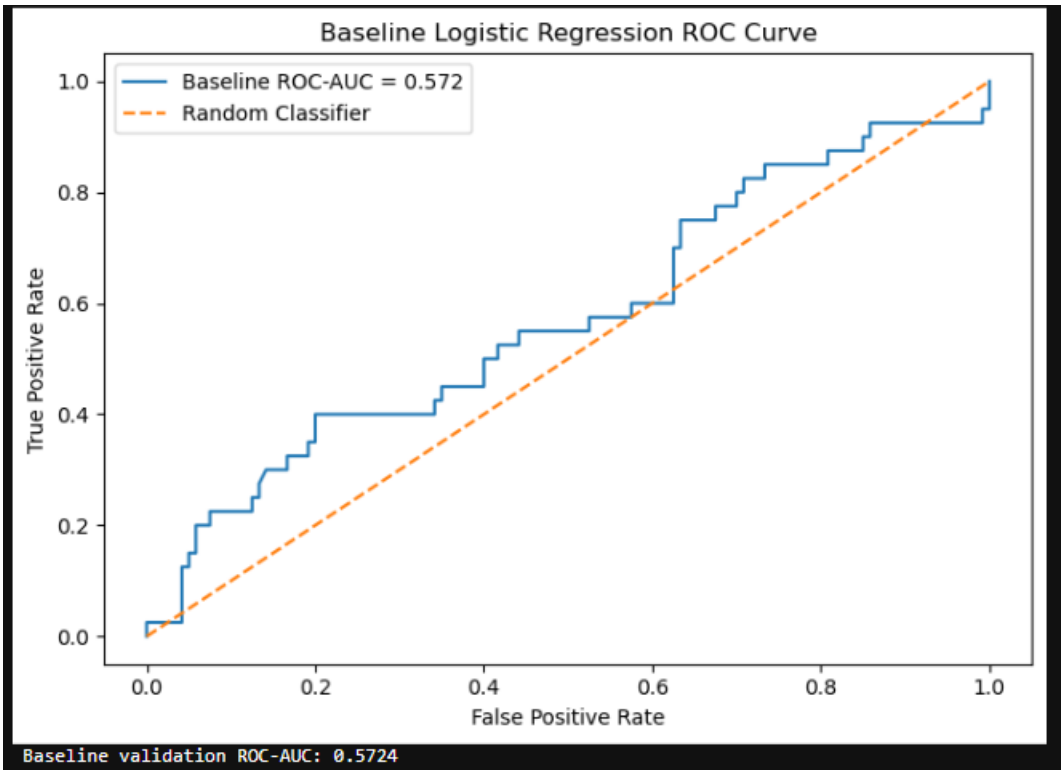


Figure 20: ROC curve for the baseline Logistic Regression model on validation data

The validation ROC-AUC was 0.5724.

The score should not be interpreted as strong predictive performance. Instead, it justified further investigation because probability ranking was slightly better than random even though the threshold-based classifier was ineffective.

8.7 Probability Distribution by Actual Class

The predicted probabilities were then compared according to the actual validation class.

Actual class	Count	Mean	Median	Standard deviation	Minimum	Maximum
Non-Bestseller	120	0.2424	0.2239	0.0559	0.1505	0.3889
Bestseller	40	0.2575	0.2486	0.0681	0.1317	0.4338

Table 21: Baseline predicted-probability summary by actual validation class

Actual bestsellers received a slightly higher average predicted probability of **0.2575**, compared with **0.2424** for non-bestsellers. Their median probability was also slightly higher at **0.2486**, compared with **0.2239**.

However, the difference was small and the probability ranges overlapped substantially.

This provides an important interpretation of the ROC-AUC result. The model was assigning somewhat higher probabilities to actual bestsellers on average, but the classes were not clearly separated.

Most importantly, both groups were concentrated below the original threshold of 0.50.

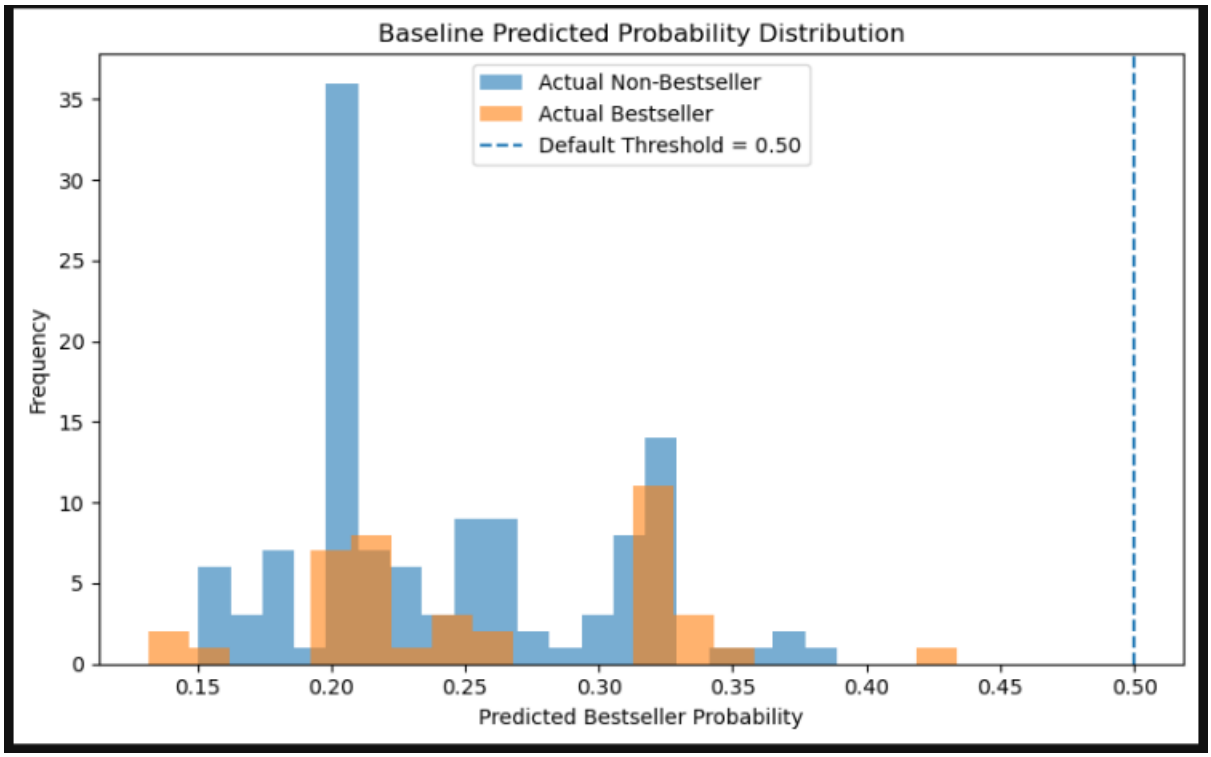


Figure 21: Distribution of baseline predicted bestseller probabilities by actual validation class

The vertical threshold line at 0.50 visually demonstrates why the baseline model produced no positive classifications. The probability distributions also show considerable overlap, indicating that changing the threshold could improve recall but would inevitably introduce false positives.

8.8 Selection of Evaluation Criteria for Model Improvements

The baseline findings demonstrated that **accuracy was unsuitable as the primary model-selection criterion**.

A model could achieve 75% accuracy while identifying zero bestsellers. Consequently, subsequent model configurations were evaluated primarily using the **F1-score**.

F1-score was selected because both false positives and false negatives were relevant to the classification objective. In the publishing scenario:

- a **false negative** represents a potential bestseller that the model fails to identify; while
- a **false positive** represents a non-bestseller that is incorrectly flagged as a potential bestseller.

Using F1-score therefore provided a compromise between precision and recall.

Balanced accuracy and **ROC-AUC** were retained as secondary measures. Balanced accuracy ensured that performance on the majority non-bestseller class did not dominate the evaluation, while ROC-AUC assessed the ranking quality of the predicted probabilities independently of a specific threshold.

Importantly, this evaluation criterion was established **before comparing the later model alternatives**. This avoided changing the preferred metric after observing which model performed better.

This model-selection strategy was established before the improvement experiments began.

8.9 Baseline Model Diagnosis

The baseline analysis identified three main problems.

Baseline finding	Evidence	Implication
Default threshold too high for predicted probability range	Maximum probability = 0.4338	Investigate lower classification thresholds
Minority class ignored	Recall = 0.0000; TP = 0	Model requires greater sensitivity to bestseller observations
Limited class discrimination	ROC-AUC = 0.5724; probability distributions overlap	Threshold adjustment alone may not completely solve the modelling problem

Table 22: Baseline model weaknesses and resulting improvement strategy

These findings provided a clear justification for model improvement rather than changing parameters arbitrarily.

Two initial controlled interventions were therefore required:

1. **classification-threshold optimisation** using validation data; and
2. **class-weighted Logistic Regression**, which would modify the training objective so that errors involving minority-class bestseller observations received greater importance.

Further improvements, including early stopping and feature-set comparison, could then be evaluated against the same predefined validation criteria.

The final test data remained untouched throughout this process.

8.10 Section Summary

The baseline Logistic Regression model achieved **75.00% validation accuracy**, but this figure was misleading because the classifier predicted all 160 validation observations as non-bestsellers. Its accuracy and balanced accuracy were identical to a naïve majority-class benchmark, while bestseller precision, recall and F1-score were all zero. The confusion matrix confirmed that all 40 actual bestsellers were missed. However, a ROC-AUC of **0.5724** and slightly higher average probabilities among actual bestsellers indicated that the predicted probabilities contained limited ranking information. The baseline evaluation

therefore justified further model development. F1-score was selected as the primary validation criterion, with balanced accuracy and ROC-AUC used as supporting metrics.

9. MODEL IMPROVEMENT AND FINAL MODEL SELECTION

The baseline evaluation demonstrated that the initial Logistic Regression classifier required improvement. Although it achieved 75.00% validation accuracy, it failed to identify any bestsellers at the default threshold of 0.50. Model improvement was therefore conducted as a sequence of **controlled experiments**, with each change motivated by a specific weakness identified during validation.

Importantly, the **test partition remained untouched** throughout this stage. Model settings were selected using training and validation data only so that the final test evaluation could remain an independent assessment of generalisation performance.

As established in Section 8, **F1-score was defined in advance as the primary model-selection metric** because the bestseller class was under-represented and both false-positive and false-negative classifications were relevant. Balanced accuracy and ROC-AUC were retained as secondary measures.

9.1 Classification-Threshold Optimisation

The first experiment investigated whether the poor baseline classification performance was partly caused by the conventional **0.50 probability threshold**.

The baseline probability analysis had already shown that the highest validation probability was only 0.4338. Therefore, retaining a threshold of 0.50 guaranteed that every validation observation would be classified as a non-bestseller.

A threshold search was conducted between **0.10 and 0.50 in increments of 0.01**. At every threshold, precision, recall, balanced accuracy and F1-score were recalculated. The threshold producing the highest validation F1-score was selected, with balanced accuracy used as a secondary criterion where necessary.

The best baseline threshold was:

$$Threshold = 0.19$$

At this threshold, the validation results were:

Metric	Threshold 0.50	Threshold 0.19
Accuracy	0.7500	0.3375
Balanced Accuracy	0.5000	0.5333
Precision	0.0000	0.2643
Recall	0.0000	0.9250
Specificity	1.0000	0.1417
F1-Score	0.0000	0.4111
ROC-AUC	0.5724	0.5724

Table 23: Effect of validation-based threshold optimisation on the baseline Logistic Regression model

The threshold search identified 0.19 as the strongest threshold according to the preselected F1 criterion, producing an F1-score of **0.4111**, precision of **0.2643** and recall of **0.9250**.

This represented a substantial improvement in the model's ability to detect bestsellers. At the 0.19 threshold, **37 of the 40 actual bestsellers** were identified. However, this improvement came at a considerable cost: specificity decreased to only **0.1417**, with 103 non-bestsellers incorrectly classified as bestsellers.

Therefore, threshold adjustment demonstrated that the underlying probabilities contained useful information, but it did not completely solve the classification problem.

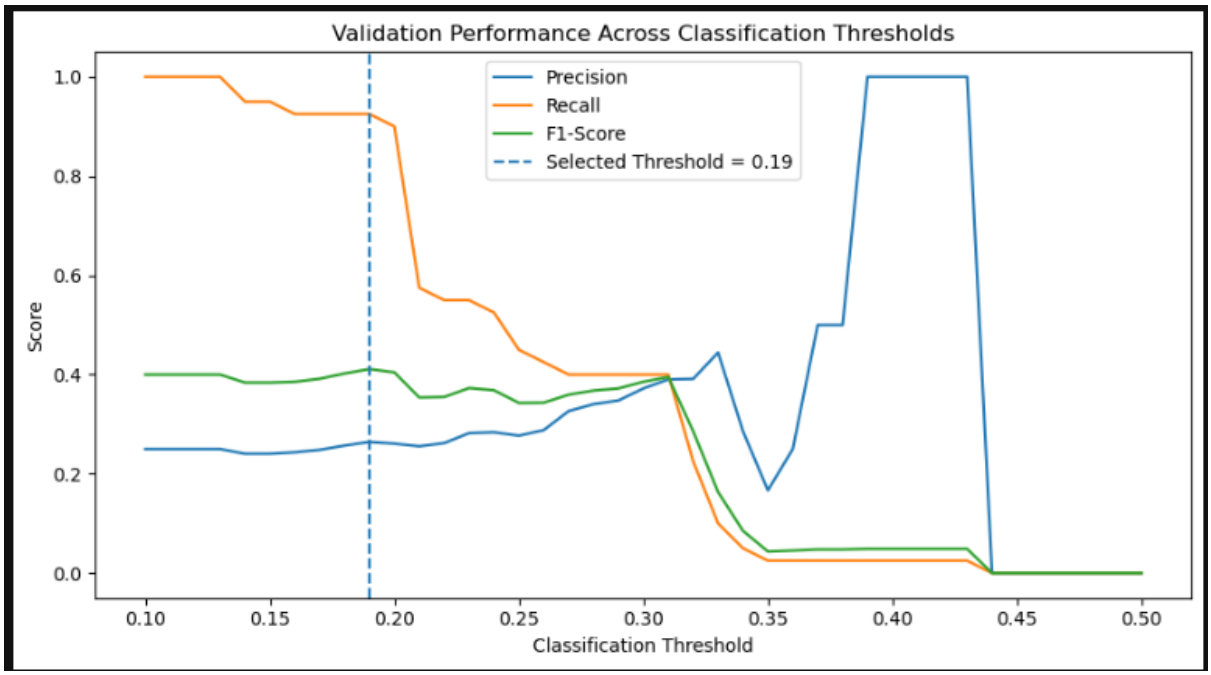


Figure 22: Validation precision, recall and F1-score across alternative classification thresholds

The figure is useful because it visually demonstrates the trade-off created by changing the classification threshold. Lower thresholds increased recall, while higher thresholds became more conservative.

9.2 Introduction of Class Weighting

The second experiment addressed the **3:1 imbalance** between non-bestseller and bestseller observations.

Under ordinary binary cross-entropy, each observation contributes equally to the loss function. Because the training dataset contained substantially more non-bestsellers than bestsellers, majority-class observations could exert greater aggregate influence during optimisation. Class weighting was therefore introduced to increase the influence of errors involving the minority bestseller class (Géron, 2022).

Balanced class weights were calculated from the 749 training observations:

- Non-Bestseller observations = **561**

- Bestseller observations = **188**

The resulting weights were:

$$w_0 = 0.6676$$

for non-bestsellers and:

$$w_1 = 1.9920$$

for bestsellers.

These values were calculated directly from the training distribution.

Class	Training observations	Class weight
Non-Bestseller	561	0.6676
Bestseller	188	1.9920

Table 24: Balanced class weights used during weighted Logistic Regression training

The manually implemented binary cross-entropy and gradient calculations were then extended to incorporate these weights. Bestseller errors consequently received approximately three times the weight assigned to non-bestseller errors.

This was still a **from-scratch implementation**. The model did not use a pre-built Logistic Regression estimator; the weighted loss and weighted gradients were explicitly implemented within the NumPy training procedure.

9.3 Initial Class-Weighted Model

The class-weighted model retained the original learning rate of **0.05** and maximum of **5,000 iterations** so that the effect of weighting could be examined without simultaneously changing multiple optimisation parameters.

Weighted training loss decreased from:

$$0.693147$$

to:

$$0.673968$$

while final validation loss was:

$$0.679580$$

and the learned intercept was 0.586209.

At the default threshold of 0.50, the class-weighted model achieved:

Metric	Class-weighted model
Accuracy	0.5562
Balanced Accuracy	0.5375
Precision	0.2817
Recall	0.5000
Specificity	0.5750
F1-Score	0.3604
ROC-AUC	0.5768

Table 25: Validation performance of the initial class-weighted Logistic Regression model

Compared with the original baseline model, class weighting produced a much more balanced classifier. Bestseller recall increased from 0.0000 to **0.5000**, while specificity remained above 0.50 at **0.5750**. ROC-AUC also increased slightly from 0.5724 to **0.5768**.

However, the weighted training-loss graph revealed an additional issue. Validation loss reached its minimum considerably before the final training iteration and subsequently began to rise while training loss continued to decline. This pattern suggested that continuing optimisation for all 5,000 iterations was beginning to fit the training observations more closely without improving validation performance.

The next experiment therefore introduced **early stopping**.

9.4 Validation-Based Early Stopping

Early stopping was added to retain the parameter values associated with the lowest validation loss rather than automatically using the weights produced by the final iteration.

The early-stopping configuration was:

Parameter	Setting
Learning rate	0.05
Maximum iterations	5,000
Patience	250 iterations
Minimum improvement (*min_delta*)	1×10^{-6}

Table 26: Early-stopping configuration for the class-weighted model

The algorithm monitored validation loss after every iteration. Whenever validation loss improved sufficiently, the current weights and intercept were saved. Training terminated when no meaningful improvement occurred for 250 consecutive iterations.

The strongest validation loss occurred at **iteration 618**. Training stopped after **868 iterations**, after which the model retained the weights from iteration 618 rather than the later parameters. The lowest validation loss was **0.673134**.

This reduced unnecessary training and provided a direct response to the overfitting pattern identified in the earlier weighted loss curve.

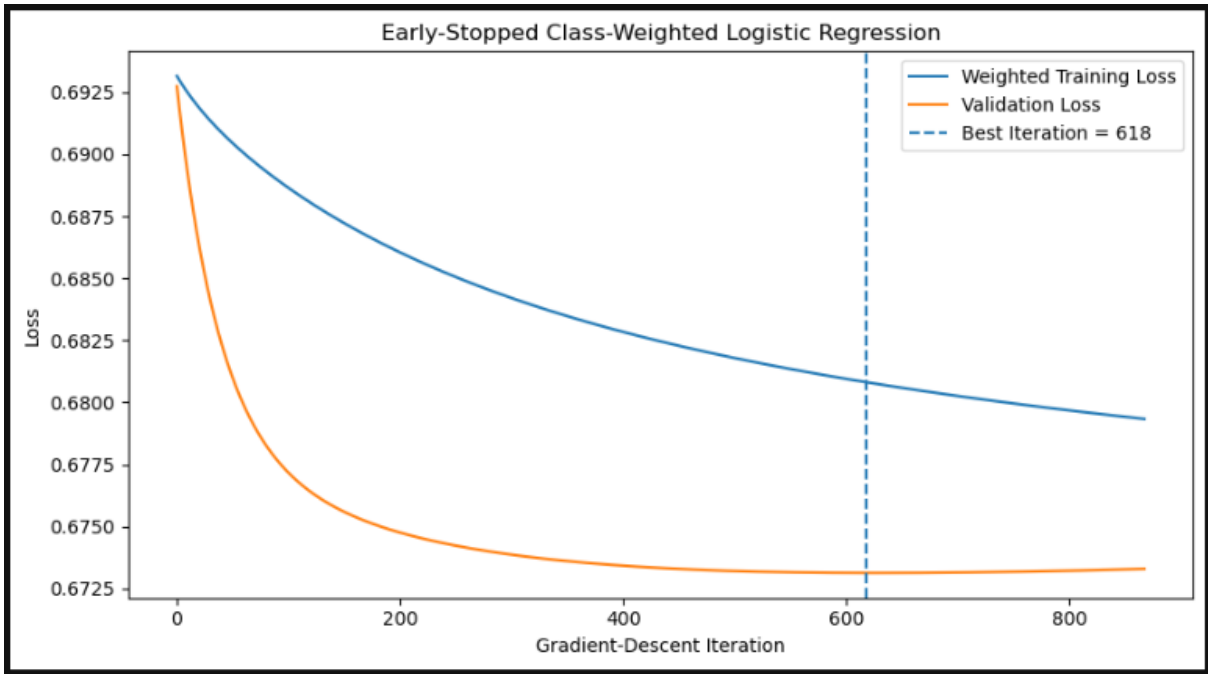


Figure 23: Training and validation loss for the class-weighted Logistic Regression model with early stopping

The dashed vertical line in this figure marks the iteration associated with the lowest validation loss, making it one of the strongest pieces of evidence for the early-stopping decision.

9.5 Threshold Optimisation After Early Stopping

After early stopping had determined the preferred model parameters, the validation threshold was optimised again rather than automatically retaining 0.50.

The best threshold for the early-stopped weighted model was:

$$Threshold = 0.44$$

At this threshold, the model achieved:

Metric	**Early-stopped weighted model**
Accuracy	0.3750
Balanced Accuracy	0.5500
Precision	0.2727
Recall	0.9000
Specificity	0.2000

Metric	Early-stopped weighted model
F1-Score	**0.4186**
ROC-AUC	0.5864

Table 27: Validation performance of the early-stopped class-weighted model at the optimised 0.44 threshold

The threshold search produced an F1-score of **0.4186**, balanced accuracy of 0.5500, precision of 0.2727 and recall of 0.9000.

The model correctly identified **36 of the 40 actual bestsellers**, while four were missed. However, the high recall came with low specificity because many non-bestsellers were also flagged as potential bestsellers.

This reflects an important modelling trade-off. The selected threshold was intentionally more sensitive to bestseller observations rather than optimised for raw accuracy.

9.6 Comparison of Model-Development Experiments

The four main configurations evaluated up to this point were compared using the same validation partition.

Model	Accuracy	Balanced Accuracy	Precision	Recall	F1-Score	ROC-AUC
Baseline, threshold 0.50	0.7500	0.5000	0.0000	0.000	0.0000	0.5724
Baseline, threshold 0.19	0.3375	0.5333	0.2643	**0.925**	0.4111	0.5724
Class-weighted, threshold 0.50	0.5562	0.5375	**0.2817**	0.500	0.3604	0.5768
Early-weighted, threshold 0.44	0.3750	**0.5500**	0.2727	0.900	**0.4186**	**0.5864**

Table 28: Validation comparison of the main Logistic Regression development experiments

The comparison shows that no single modification improved every metric simultaneously. The threshold-adjusted baseline achieved the highest recall, while the default class-weighted model produced the highest precision among these four configurations. However, the **early-stopped weighted model achieved the highest F1-score of 0.4186**, which was the predefined primary criterion. It also produced the strongest ROC-AUC among these configurations.

This progression illustrates why model development should not depend on a single accuracy value. Improvements were judged according to the classification objective rather than simply selecting the configuration with the highest proportion of overall correct predictions.

9.7 Reduced-Feature Experiment

The exploratory analysis had shown that Author_Rating produced the strongest univariate categorical relationship with bestseller status. A reduced-feature model was therefore developed to determine whether the weaker predictors added sufficient value to justify retaining the complete feature set.

Only Author_Rating was retained in the reduced model. After one-hot encoding with one reference category removed, it produced three processed predictors:

- Author_Rating_Famous
- Author_Rating_Intermediate
- Author_Rating_Novice

The reduced model used the **same class-weighted early-stopping algorithm**, ensuring that the feature-set comparison remained controlled.

Its validation-selected threshold was **0.46**, producing:

Metric	Reduced Author_Rating model
Accuracy	0.6312
Balanced Accuracy	0.5958
Precision	0.3443
Recall	0.5250
Specificity	0.6667
F1-Score	0.4158
ROC-AUC	0.5971

Table 29: Validation performance of the reduced Author_Rating Logistic Regression model

The threshold optimisation selected 0.46 with an F1-score of **0.4158**, balanced accuracy of 0.5958, precision of 0.3443 and recall of 0.5250.

The reduced model performed particularly well in terms of balanced accuracy, specificity, precision and ROC-AUC. This reinforced the earlier EDA finding that *Author_Rating* contained a substantial proportion of the predictive signal available in the dataset.

9.8 Full Versus Reduced Feature Set

The final feature-selection decision required comparison of the full and reduced models using the previously defined selection criterion.

Model	Processed features	Threshold	Balanced Accuracy	Precision	Recall	F1-Score	ROC-AUC
Full Early-Weighted Model	16	0.44	0.5500	0.2727	**0.9000**	**0.4186**	0.5864
Reduced Author_Rating Model	3	0.46	**0.5958**	**0.3443**	0.5250	0.4158	**0.5971**

Table 30: Validation comparison of the full and reduced Logistic Regression feature sets

The reduced model produced higher balanced accuracy, precision and ROC-AUC. However, the **full model achieved the slightly higher F1-score**, 0.4186 compared with 0.4158, and substantially greater recall of 0.9000 compared with 0.5250.

This created a genuine modelling trade-off rather than an obvious overall winner.

Because **F1-score had been defined before these results were examined as the primary model-selection metric**, the full model was retained. Changing to balanced accuracy or ROC-AUC only after observing that the reduced model performed better on those metrics would have represented a post-hoc change in the model-selection criterion.

Maintaining the original criterion therefore provided a more consistent and defensible model-selection process.

	Model	Processed Features	Threshold	Balanced Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Full Early-Weighted Model	16	0.44	0.5500	0.2727	0.900	0.4186	0.5864
1	Reduced Author_Rating Model	3	0.46	0.5958	0.3443	0.525	0.4158	0.5971

Figure 24: Validation comparison of the full early-weighted and reduced Author_Rating models

9.9 Final Model Selection

The final model selected for test-set evaluation was therefore the **full early-stopped class-weighted Logistic Regression model**.

Its final configuration consisted of:

Component	Final selection
Raw predictor set	Full six leakage-controlled predictors
Processed predictor count	16
Loss function	Class-weighted binary cross-entropy
Non-bestseller class weight	0.6676
Bestseller class weight	1.9920
Optimisation	Batch gradient descent
Learning rate	0.05
Maximum iterations	5,000
Early-stopping patience	250
Best retained iteration	618
Classification threshold	**0.44**
Primary selection metric	F1-score
Validation F1-score	**0.4186**

Table 31: Final Logistic Regression model configuration selected before test evaluation

The full model was selected because it achieved the highest F1-score among the evaluated feature-set alternatives and because F1 had been defined in advance as the primary criterion. All model-selection decisions were completed before accessing the test partition.

This is methodologically important. The unseen test data were not used to decide the threshold, class weights, early-stopping point, feature set or preferred model. The next stage could therefore evaluate the selected configuration without making any further changes.

9.10 Section Summary

The model-improvement stage used validation data to address weaknesses identified in the baseline classifier. Lowering the baseline classification threshold from 0.50 to 0.19 increased recall to 0.9250 and F1-score to 0.4111, but resulted in very low specificity. Class weighting was then introduced to increase the contribution of minority bestseller observations during optimisation. Early stopping further improved the training process by retaining the parameter values associated with the minimum validation loss at iteration 618. After threshold optimisation, the early-stopped weighted model achieved the highest validation F1-score of **0.4186** at a threshold of **0.44**. A reduced model containing only *Author_Rating* achieved stronger balanced accuracy and ROC-AUC but a slightly lower F1-score of 0.4158. Because F1-score had been selected in advance as the primary criterion, the full 16-feature early-weighted model was retained for final evaluation.

10. FINAL TEST EVALUATION

After the full early-stopped class-weighted Logistic Regression model had been selected using validation performance, it was evaluated **once on the previously untouched test partition**. The test set contained **161 books** and had not been used to determine the feature set, preprocessing parameters, classification threshold, class weights, early-stopping point or model-selection decision. This separation provided an independent assessment of the selected model's behaviour on unseen observations.

The final classification threshold remained fixed at **0.44**, as selected during validation. No further parameter or threshold changes were made after the test results were observed.

10.1 Final Test Predictions

Using the fixed threshold of 0.44, the final model classified:

- **29 books as non-bestsellers**
- **132 books as bestsellers**

These predictions were generated directly from the selected model without retraining or additional test-based optimisation.

The large number of positive predictions was consistent with the model-development objective established during validation: the selected configuration prioritised **high bestseller recall**, accepting more false-positive predictions in order to reduce the number of actual bestsellers that were missed.

10.2 Final Test Performance

The selected Logistic Regression model achieved the following results on the 161 unseen test observations:

Metric	Final test score
Accuracy	0.3789
Balanced Accuracy	0.5533
Precision	0.2727
Recall	**0.9000**
Specificity	0.2066
F1-Score	**0.4186**
ROC-AUC	**0.6728**

Table 32: Final performance of the selected Logistic Regression model on the unseen test partition

These values were produced by the final test evaluation.

The model retained the **0.4186 F1-score** achieved during validation. More importantly, recall remained at **0.9000**, showing that the classifier continued to identify 90% of actual bestsellers on observations that had not participated in model development.

Overall accuracy remained relatively low at **37.89%**. This is a direct consequence of the model's high-sensitivity decision rule. Many non-bestsellers were deliberately classified as

potential bestsellers in exchange for reducing missed bestseller cases. For this reason, the final results should continue to be interpreted using class-specific measures rather than accuracy in isolation (Géron, 2022).

FINAL TEST PERFORMANCE

	Metric	Score
0	Accuracy	0.3789
1	Balanced Accuracy	0.5533
2	Precision	0.2727
3	Recall	0.9000
4	Specificity	0.2066
5	F1-Score	0.4186
6	ROC-AUC	0.6728

CONFUSION MATRIX COUNTS

TN: 25
FP: 96
FN: 4
TP: 36

Figure 25: Final evaluation metrics for the selected Logistic Regression model on unseen test data

10.3 Final Test Confusion Matrix

The final confusion matrix provides a more detailed explanation of the test metrics.

The model produced:

- **True Negatives (TN): 25**
- **False Positives (FP): 96**
- **False Negatives (FN): 4**
- **True Positives (TP): 36**

The test partition contained **40 actual bestsellers**. Of these, 36 were successfully identified and only four were missed:

$$Recall = \frac{36}{36 + 4} = 0.9000$$

However, of the **121 actual non-bestsellers**, only 25 were correctly classified, while 96 were incorrectly flagged as bestsellers:

$$Specificity = \frac{25}{25 + 96} = 0.2066$$

Similarly, the model made 132 positive predictions, but only 36 of these were genuine bestsellers:

$$Precision = \frac{36}{36 + 96} = 0.2727$$

These confusion-matrix counts are reported directly by the final test evaluation.

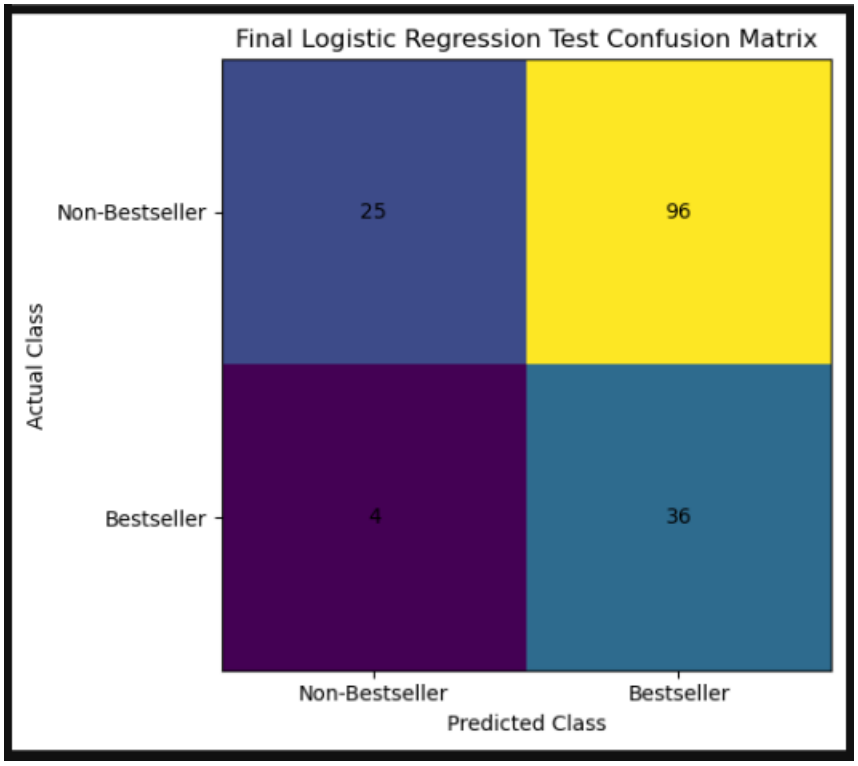


Figure 26: Confusion matrix of the selected Logistic Regression model on the unseen test set

The confusion matrix demonstrates the model's central trade-off clearly. It was effective at reducing **false negatives**, but produced a substantial number of **false positives**.

In the publisher scenario, this means that the model would be more suitable for an **initial screening process** in which potentially successful books are shortlisted for further assessment. It would be less suitable as the sole basis for expensive acquisition or marketing decisions because approximately three quarters of its positive predictions were false positives.

10.4 ROC-AUC on Unseen Data

The ROC curve was also calculated using the final test probabilities.

The selected model achieved:

$$ROC\text{-}AUC = 0.6728$$

ROC-AUC measures probability-ranking performance across all possible classification thresholds rather than evaluating only the selected 0.44 decision boundary. A value above 0.50 indicates that the model possesses greater ranking ability than a random classifier (James et al., 2023).

The test ROC-AUC of **0.6728** therefore indicates that the model's probability estimates contained useful discriminatory information, even though considerable overlap remained between bestseller and non-bestseller observations.

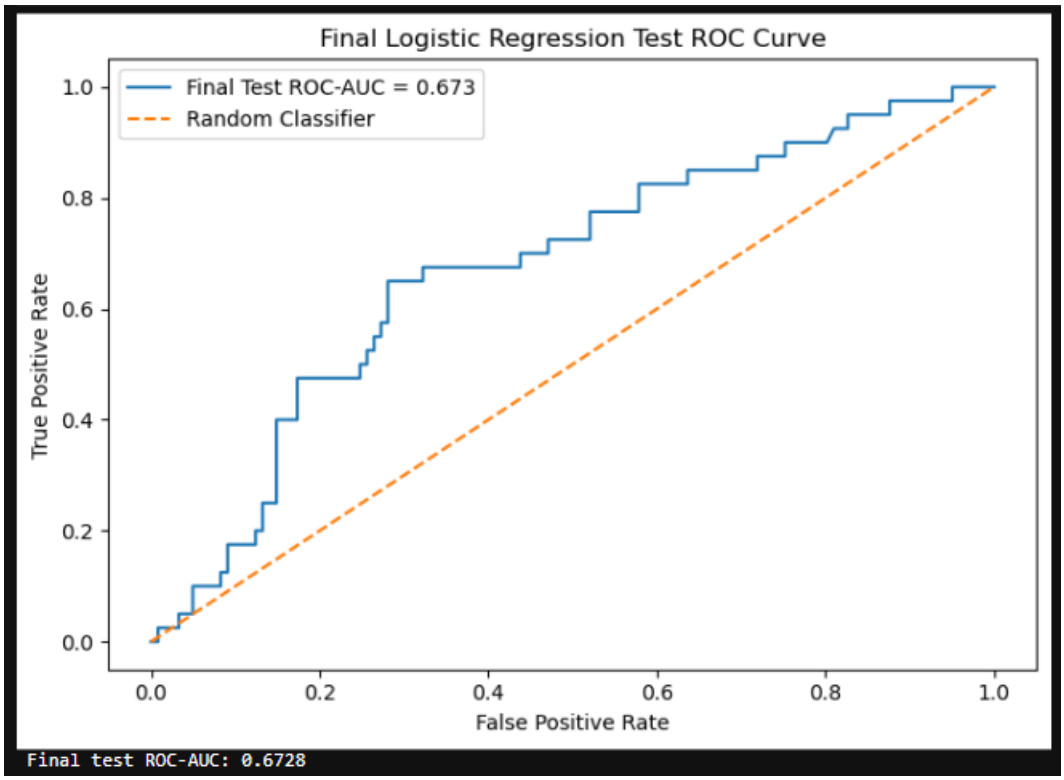


Figure 27: ROC curve of the selected Logistic Regression model on the unseen test partition

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The final test ROC-AUC was 0.6728.

The ROC-AUC result is particularly useful because the final threshold was intentionally configured for high recall. Unlike accuracy and the confusion matrix, ROC-AUC evaluates the underlying probability ordering independently of this particular classification threshold.

10.5 Validation Versus Test Performance

The final model's validation and test results were compared to determine whether the behaviour observed during model development carried forward to unseen data.

Metric	Validation	Test	Difference
Accuracy	0.3750	0.3789	+0.0039
Balanced Accuracy	0.5500	0.5533	+0.0033
Precision	0.2727	0.2727	0.0000
Recall	0.9000	0.9000	0.0000
Specificity	0.2000	0.2066	+0.0066
F1-Score	0.4186	0.4186	0.0000
ROC-AUC	0.5864	0.6728	+0.0865

Table 33: Comparison of validation and final test performance for the selected Logistic Regression model

The comparison shows considerable consistency in the **threshold-dependent metrics**. Precision, recall and F1-score were identical to four decimal places across validation and test data. Accuracy, balanced accuracy and specificity changed only slightly.

This consistency is useful because it suggests that the classification behaviour selected during validation did not deteriorate substantially when applied to the test partition.

The largest change occurred in ROC-AUC, which increased from **0.5864 during validation to 0.6728 during testing**, an absolute difference of approximately **0.0865**. This indicates that the ordering of predicted probabilities separated the two classes more effectively within the unseen test sample than within the validation sample.

However, this improvement should not be treated as evidence that the model will necessarily achieve the same ROC-AUC on every future sample. Both validation and test partitions are finite samples from the same dataset, and performance can vary between samples (James et al., 2023).

	Metric	Validation	Test	Difference
0	Accuracy	0.3750	0.3789	0.0039
1	Balanced Accuracy	0.5500	0.5533	0.0033
2	Precision	0.2727	0.2727	0.0000
3	Recall	0.9000	0.9000	0.0000
4	Specificity	0.2000	0.2066	0.0066
5	F1-Score	0.4186	0.4186	0.0000
6	ROC-AUC	0.5864	0.6728	0.0865

Figure 28: Comparison of validation and final test performance

10.6 Practical Interpretation of Final Performance

The results indicate that the final model is considerably better at **finding potential bestsellers than rejecting non-bestsellers**.

Its most important strength was recall:

$$Recall = 0.9000$$

Only **4 of the 40 actual bestsellers** were missed.

Its main limitation was the large number of false positives. The model incorrectly classified **96 of the 121 actual non-bestsellers** as bestsellers. Precision was consequently only 0.2727 and specificity only 0.2066.

This means the model should not be interpreted as a system capable of confidently deciding whether a book **will** become a bestseller. Instead, it is more appropriately interpreted as a **high-recall screening model**.

A possible publisher workflow would therefore be:

Book submitted → Logistic Regression screening → potential bestseller shortlist → additional commercial/editorial assessment

rather than:

Book submitted → Logistic Regression prediction → automatic acquisition decision

This distinction is important because predictive models should be applied according to the types and costs of errors they produce. A classifier with high recall but low precision may be useful when missing a positive case is considered more costly than investigating additional false-positive cases (Géron, 2022).

10.7 Final Model Performance Assessment

The final evaluation should not be presented as evidence of strong overall bestseller prediction.

The model achieved:

- **high recall (0.9000);**
- **an F1-score of 0.4186;**
- **balanced accuracy slightly above random classification at 0.5533;**
- **ROC-AUC of 0.6728;**
- but **low precision (0.2727)** and **low specificity (0.2066)**.

The results therefore demonstrate that the available pre-outcome variables contained **some useful predictive signal**, but not enough information for highly reliable separation of bestsellers and non-bestsellers.

This outcome is also consistent with the EDA. Most available predictors showed weak standalone relationships with bestseller status, meaning that exceptionally strong classification performance would have been unexpected.

The value of the modelling process is therefore not that an artificially high performance score was produced, but that the final evaluation honestly identifies **where the model succeeds, where it fails and how it could realistically be used**.

10.8 Section Summary

The final selected Logistic Regression model was evaluated once on the untouched test partition of 161 books using the validation-selected threshold of 0.44. It achieved an F1-score of **0.4186**, balanced accuracy of **0.5533** and ROC-AUC of **0.6728**. Recall remained high at **0.9000**, with 36 of the 40 actual bestsellers correctly identified. However, only 25 of the 121 non-bestsellers were correctly rejected, resulting in 96 false positives, precision of **0.2727** and specificity of **0.2066**. Validation and test F1-score, recall and precision were identical to four decimal places, indicating stable threshold-based behaviour across the two partitions. Overall, the model is best interpreted as an exploratory high-recall screening tool rather than a definitive bestseller forecasting system.

MODEL INTERPRETATION AND ERROR ANALYSIS

After evaluating the final classifier, the analysis moved beyond overall performance metrics to examine **how the model formed its predictions and where its errors occurred**. Logistic Regression is useful in this respect because each fitted coefficient indicates the direction and relative magnitude of the relationship between a predictor and the estimated log-odds of the positive class, while exponentiating a coefficient produces an odds ratio (James et al., 2023).

Interpretation was conducted cautiously. The model was trained using **class-weighted loss**, and the data were observational rather than experimental. The coefficients and odds ratios are therefore interpreted as **associations within the fitted predictive model**, not as causal effects or population-level inferential estimates.

10.9 Categorical Reference Levels

Because categorical predictors were transformed using one-hot encoding with drop="first", one category from each categorical variable was omitted and used as the reference group.

The final reference categories were:

Predictor	Reference category
Author_Rating	Excellent
genre	children
Publisher	Amazon Digital Services, Inc.
language_group	English

Table 34: Reference categories used when interpreting Logistic Regression coefficients

All dummy-variable coefficients must be interpreted **relative to these reference categories**, while the remaining model features are held constant.

For example, Author_Rating_Intermediate does not represent the absolute effect of being rated Intermediate. Instead, its coefficient describes the fitted difference between an Intermediate-rated author and the omitted **Excellent** reference category.

10.10 Interpretation of Coefficients and Odds Ratios

For Logistic Regression, the model estimates:

$$log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

where p represents the estimated probability of bestseller classification.

A positive coefficient indicates higher estimated bestseller log-odds as that feature increases or relative to its reference category, while a negative coefficient indicates lower estimated log-odds.

Each coefficient was exponentiated to produce an odds ratio:

$$e^{\beta}$$

An odds ratio above 1 indicates higher fitted bestseller odds, whereas an odds ratio below 1 indicates lower fitted odds, subject to the relevant reference condition (James et al., 2023).

Because *Publishing Year* and *sale price* were standardised before model fitting, their coefficients describe changes associated with a **one-standard-deviation increase**, rather than a one-year or one-currency-unit increase.

10.11 Strong Model Effects

The coefficients were ranked according to their absolute magnitude to identify the variables that had the greatest influence within the fitted model.

Feature	Coefficient	Odds ratio	Direction
Author_Rating_Intermediate	-0.3718	0.6895	Lower
Author_Rating_Famous	-0.1959	0.8221	Lower
Publisher_Random House LLC	+0.1844	1.2026	Higher
Publisher_Penguin Group (USA) LLC	-0.1702	0.8435	Lower
genre_nonfiction	+0.1051	1.1108	Higher
Author_Rating_Novice	-0.0998	0.9050	Lower
Publisher_Macmillan	-0.0923	0.9118	Lower
Publisher_Hachette Book Group	+0.0840	1.0877	Higher

Table 35: Strongest coefficients and odds ratios in the final Logistic Regression model

The strongest coefficient was associated with *Author_Rating_Intermediate*. Because **Excellent** was the reference category, its coefficient of **-0.3718** and odds ratio of **0.6895** indicate that Intermediate-rated authors had lower fitted bestseller odds than Excellent-rated authors, with the remaining model variables held constant.

Author_Rating_Famous also had a negative coefficient of **-0.1959**, corresponding to an odds ratio of **0.8221**. The *Novice* category similarly produced an odds ratio below one. These results are broadly consistent with the earlier EDA, where books in the Excellent author-rating group displayed the highest observed bestseller rate.

Publisher effects were smaller. *Random House LLC* had a positive coefficient of **0.1844** and an odds ratio of **1.2026** relative to the reference publisher, Amazon Digital Services, Inc. In contrast, *Penguin Group (USA) LLC* had an odds ratio of **0.8435**.

The nonfiction genre coefficient was also positive, with an odds ratio of **1.1108** relative to the children's-book reference category. However, the children's category contained only 15 observations, meaning this comparison should be interpreted cautiously.

The remaining coefficients were relatively close to zero. For example, standardised *sale price* had a coefficient of **0.0349**, while standardised *Publishing Year* had a coefficient of **-0.0314**. These small effects are consistent with the weak numerical relationships observed during EDA.

TOP 10 HIGHEST-PROBABILITY FALSE POSITIVES

	Book Name	Author	Author_Rating	genre	sale price	Publisher	units sold	Predicted Probability
1053	Rosencrantz and Guildenstern are Dead	Tom Stoppard, Henry Popkin	Excellent	nonfiction	12.59	Random House LLC	106	0.6389
456	A Monster Calls	Patrick Ness, Jim Kay, Siobhan Dowd	Excellent	fiction	10.65	Random House LLC	4671	0.5991
833	Ð—Ð°Ð¿Ð¸ÑŠÐºÐ¸ Ð¿Ð¾ Ð¿Ñ¾Ð´Ð¿Ð¾Ð»ÑŒÑŒÑ	Fyodor Dostoyevsky, Andrew R. MacAndrew, Ben M...	Excellent	fiction	7.99	Random House LLC	2889	0.5983
942	Lover Enshrined, part one	J.R. Ward	Excellent	fiction	8.99	Random House LLC	4320	0.5953
981	Twelve Years a Slave	Solomon Northup	Excellent	fiction	4.99	Random House LLC	4240	0.5917
1045	Full Dark, No Stars	Stephen King	Excellent	fiction	5.99	Random House LLC	107	0.5882
297	Persepolis	Marjane Satrapi, Mattias Ripa	Excellent	fiction	4.99	Random House LLC	298	0.5862
568	é□"å¾·ç¶" [dÃ o dÃ© jÃ«ng]	Lao Tzu, Gia-Fu Feng, Jane English, Chungliang...	Excellent	fiction	4.59	Penguin Group (USA) LLC	3915	0.5857
953	The Fall of Hyperion	Dan Simmons	Excellent	nonfiction	5.83	Simon and Schuster Digital Sales Inc	4280	0.5853
879	The Murder of Roger Ackroyd	Agatha Christie	Excellent	nonfiction	7.52	Amazon Digital Services, Inc.	4440	0.5849

Figure 29: Direction and magnitude of coefficients in the final Logistic Regression model

10.12 Interpretation Caution

Although odds ratios make the Logistic Regression model easier to interpret, they should not be treated as evidence that the corresponding book characteristics **cause** bestseller performance.

There are three important reasons for this.

First, the analysis is based on observational data and does not experimentally manipulate publisher, genre, author rating or pricing. Second, the final model was trained using class-weighted binary cross-entropy so that bestseller observations received greater importance during optimisation. Third, the target itself represents a dataset-specific operational definition based on the upper quartile of observed unit sales.

The odds ratios should therefore be interpreted as **model-based predictive associations within this dataset**, rather than causal effects or population-level inferential odds ratios.

This distinction is particularly important when discussing variables such as publisher or author rating. A positive coefficient does not establish that changing a publisher or author classification would cause a book to become a bestseller.

10.13 Prediction Error Analysis

Model interpretation was extended by examining individual classification errors on the unseen test data.

The test results consisted of:

Prediction outcome	Count
True Positive	36
True Negative	25
False Positive	96
False Negative	4

Table 36: Distribution of final test prediction outcomes

This error distribution reinforces the finding from Section 10: the final model was configured to favour high recall. As a result, relatively few actual bestsellers were missed, but a large number of non-bestsellers were incorrectly flagged as potential bestsellers.

10.14 False-Negative Analysis

False negatives are particularly important in the publishing scenario because they represent books that satisfied the operational bestseller definition but were incorrectly rejected by the model.

Only four false negatives were produced:

Book	Author Rating	Genre	Units sold	Predicted probability
The Miserable Mill	Intermediate	fiction	46,872	0.4208
Warm Bodies	Intermediate	fiction	43,848	0.4328
The Amulet of Samarkand	Intermediate	fiction	47,736	0.4384
The Monk Who Sold His Ferrari	Intermediate	fiction	40,824	0.4386

Table 37: False-negative bestseller observations in the unseen test set

A particularly important pattern is that **all four false negatives had probabilities below, but relatively close to, the selected 0.44 classification threshold**.

The highest false-negative probability was:

$$0.4386$$

which is only:

$$0.44 - 0.4386 = 0.0014$$

below the classification boundary.

Similarly, the false-negative probabilities ranged only from **0.4208 to 0.4386**.

This means that the model did not assign extremely low bestseller probabilities to these books. Instead, they narrowly fell on the non-bestseller side of the selected threshold.

All four were also associated with the Intermediate author-rating category and the fiction genre. The strong negative coefficient associated with Author_Rating_Intermediate provides a plausible **model-based explanation** for part of this pattern, although the predictions result from all 16 processed predictors acting jointly.

FALSE NEGATIVES

	Book Name	Author	Author_Rating	genre	sale price	Publisher	units sold	Predicted Probability
632	The Miserable Mill	Lemony Snicket, Brett Helquist	Intermediate	fiction	10.99	Penguin Group (USA) LLC	46872	0.4208
646	Warm Bodies	Isaac Marion	Intermediate	fiction	3.99	Amazon Digital Services, Inc.	43848	0.4328
624	The Amulet of Samarkand	Jonathan Stroud	Intermediate	fiction	0.99	Amazon Digital Services, Inc.	47736	0.4384
658	The Monk Who Sold His Ferrari: A Fable About F...	Robin S. Sharma	Intermediate	fiction	0.99	Amazon Digital Services, Inc.	40824	0.4386

Figure 30: False-negative bestseller observations produced by the final model

10.15 False-Positive Analysis

False positives represented the largest source of error. A total of **96 non-bestsellers** were classified as bestsellers.

Unlike the false negatives, not all false-positive cases occurred immediately above the 0.44 threshold. The highest predicted probability among false positives reached:

$$0.6389$$

For example, the highest-probability false positive was *Rosencrantz and Guildenstern are Dead*, which had only **106 observed units sold** but received a predicted bestseller probability of **0.6389**.

Several of the highest-probability false positives shared characteristics that the fitted model associated with relatively stronger bestseller odds, particularly an Excellent author rating and, in several cases, the Random House LLC publisher category. This provides an indication of why the model may confidently classify some observations incorrectly: books can possess predictor combinations that resemble those associated with the positive class without ultimately achieving high observed sales.

This also illustrates an important limitation of the available predictors. Variables such as author rating, genre, publisher, language, price and publishing year cannot capture every factor that influences commercial book performance.

10.16 Probability Analysis by Prediction Outcome

The predicted probabilities were summarised separately for true positives, true negatives, false positives and false negatives.

Outcome	Count	Mean probability	Median	Minimum	Maximum

False Negative	4	0.4327	0.4356	0.4208	0.4386
False Positive	96	0.4980	0.4904	0.4401	0.6389
True Negative	25	0.4280	0.4294	0.4044	0.4393
True Positive	36	0.5224	0.5358	0.4415	0.6134

Table 38: Predicted-probability distributions across final test classification outcomes

The table provides two important findings.

First, the false negatives were strongly concentrated near the threshold. Their mean predicted probability was **0.4327**, and their maximum probability of 0.4386 was immediately below the 0.44 classification boundary.

Second, false positives were not exclusively borderline observations. Their mean probability was **0.4980**, and some received probabilities considerably above the threshold, reaching **0.6389**.

This indicates that the final model's errors cannot be explained simply by an imperfect threshold. If threshold selection were the only problem, most false positives and false negatives would be expected to cluster narrowly around 0.44. Instead, some non-bestsellers were assigned relatively strong positive probabilities.

There is therefore meaningful **overlap between the predictor profiles of bestsellers and non-bestsellers** in the available feature space.

Error Type	count	mean	median	min	max
False Negative	4	0.4327	0.4356	0.4208	0.4386
False Positive	96	0.4980	0.4904	0.4401	0.6389
True Negative	25	0.4280	0.4294	0.4044	0.4393
True Positive	36	0.5224	0.5358	0.4415	0.6134

Figure 31: Summary of predicted probabilities by final classification outcome

10.17 Relationship Between Model Interpretation and Earlier EDA

The coefficient results were broadly consistent with several exploratory findings.

Author_Rating had produced the strongest categorical relationship during the earlier statistical screening, and the final model again assigned its largest absolute coefficient to an author-rating category. Conversely, *sale price* and *Publishing Year* had shown very weak univariate relationships during EDA and subsequently received comparatively small model coefficients.

However, the error analysis also demonstrated that even the strongest available variables did not provide sufficient separation between the two classes.

This is visible in three findings:

- actual bestsellers could receive probabilities below the classification threshold;
- non-bestsellers could receive relatively high bestseller probabilities; and

- the final ROC-AUC remained moderate at 0.6728 rather than indicating strong class separation.

The final model should therefore be interpreted as capturing **limited but measurable structure** in the available data rather than discovering a simple set of characteristics that determines bestseller performance.

10.18 Practical Interpretation

From a publishing perspective, the model's behaviour is consistent with an **early screening system**.

Its high recall means that most books meeting the operational bestseller definition would be included in the positive shortlist. However, the large false-positive count means that additional evaluation would still be required before commercial resources were committed.

The model could therefore support, rather than replace, further decision-making involving factors not contained in the dataset, such as marketing strategy, distribution reach, timing, audience demand or other relevant commercial information.

The important point is that the classifier should not be interpreted as saying:

"This book will become a bestseller."

Instead, its prediction is closer to:

"Based on the limited characteristics represented in this dataset and the fitted decision threshold, this book displays a predictor profile that the model associates with the bestseller class."

That interpretation is more consistent with both the statistical meaning of Logistic Regression and the observed predictive performance.

10.19 Section Summary

Interpretation of the final Logistic Regression model showed that *Author_Rating* contributed some of the strongest fitted effects, with the Intermediate category producing the largest absolute coefficient of **-0.3718** relative to the Excellent reference group. Publisher and genre effects were generally smaller, while sale price and publishing year had coefficients close to zero. These effects are interpreted as associations within the class-weighted predictive model rather than causal relationships. Error analysis showed that the four false negatives occurred close to the selected 0.44 threshold, with probabilities ranging from **0.4208 to 0.4386**. In contrast, the 96 false positives included observations with probabilities as high as **0.6389**, showing that classification errors were not caused by threshold placement alone. The available predictor profiles overlap substantially between bestsellers and non-bestsellers, limiting reliable class separation and reinforcing the model's role as a high-recall screening tool rather than a definitive forecasting system.

11. LIMITATIONS

Although the analysis followed a structured modelling process and maintained separation between training, validation and test data, several limitations should be considered when interpreting the results. Recognising these limitations is important because predictive performance depends not only on the algorithm but also on how the target is defined, which predictors are available and how representative the source data are (James et al., 2023).

11.1 Operational Definition of Bestseller Status

The first limitation concerns the definition of the target variable. The source dataset did **not contain an official binary bestseller label**. Bestseller status was therefore operationally defined as membership in the highest-selling 25% of observations according to *units sold*.

Using the 75th percentile produced a threshold of 5,312.25 units, with 268 books classified as bestsellers and 802 as non-bestsellers. However, this means that the model predicts high sales relative to this particular dataset, rather than membership in an externally recognised bestseller list.

Consequently, a book classified as a bestseller in this analysis should not automatically be interpreted as having appeared on an industry bestseller ranking. The definition was created specifically to make the available sales data suitable for binary Logistic Regression.

The threshold is also dataset-dependent. If the distribution of book sales changed substantially in another collection, the 75th-percentile sales value would probably differ. This limits direct generalisation of the exact bestseller definition beyond the current data.

11.2 Limited Leakage-Safe Predictor Information

A second limitation is that many variables with strong potential relationships to commercial success could not appropriately be used as predictors.

Variables including *units sold*, *gross sales*, *publisher revenue* and *sales rank* described realised commercial performance and were excluded to prevent target leakage. *Book_average_rating* and *Book_ratings_count* were also excluded because they represent reader-response information generated after books have already entered the market.

Excluding these variables produced a more realistic prediction task, but it also removed substantial predictive information.

The final classifier therefore relied mainly on six pre-outcome characteristics:

Publishing Year, sale price, Author_Rating, genre, Publisher and *language_group.*

These variables alone may not capture many factors that influence commercial performance, such as marketing activity, distribution coverage, market demand, release timing or promotional exposure. The observed classification difficulty is therefore partly a limitation of the available predictor information rather than evidence that the Logistic Regression implementation failed.

11.3 Class Imbalance and Threshold Trade-Off

The target contained approximately **25% bestsellers and 75% non-bestsellers**, creating a moderately imbalanced classification problem.

The baseline model demonstrated the practical impact of this imbalance by achieving 75% accuracy while predicting no bestsellers. Class weighting, threshold optimisation and early stopping improved minority-class detection, but the final selected threshold of **0.44** created a clear trade-off.

The final test model achieved recall of **0.9000**, successfully detecting 36 of 40 actual bestsellers. However, specificity was only **0.2066**, and 96 non-bestsellers were incorrectly classified as bestsellers.

This trade-off resulted directly from selecting F1-score as the primary validation criterion and prioritising detection of the minority bestseller class. The high false-positive count is therefore an important limitation of the selected decision rule.

The model's performance should therefore not be described simply as “accurate” or “inaccurate”. Its behaviour is asymmetric: it performs substantially better at **finding bestsellers** than at **rejecting non-bestsellers**.

11.4 Small and Unequal Categorical Groups

Several categorical predictors contained substantially unequal group sizes.

For example, the cleaned genre variable contained **884 fiction books**, **171 nonfiction books** and only **15 children's books**. Similarly, the Other language group contained only nine observations, while some publisher categories were represented by very few records.

Coefficients associated with small categories should therefore be interpreted carefully.

This limitation was also visible during chi-square assumption checking, where some genre, language and publisher contingency-table cells had expected frequencies below five.

The comparatively high observed bestseller rate among children's books, for example, should therefore not be treated as strong evidence of a general market pattern. A small change in only a few observations could materially affect the rate for such a small group.

11.5 Representativeness and Generalisability

The dataset represents a particular collection of **1,070 books**, publishers, languages and historical publication periods. It should not be assumed to represent the complete contemporary publishing market.

Relationships identified within this collection may not generalise to all publishing environments.

The final test partition provided an important assessment of previously unseen observations, but those 161 books were still drawn from the **same source dataset** as the training and

validation data. The test evaluation therefore measures internal generalisation within this collection rather than performance on an independently sourced publishing dataset.

Future validation on a separate dataset from another publisher, market or time period would provide stronger evidence concerning external generalisability.

11.6 Logistic Regression Model Assumptions

A further limitation results from the structure of Logistic Regression itself.

The model represents the target log-odds using an additive linear combination of the processed predictors. This means that more complicated non-linear relationships or interactions between book characteristics may not be represented unless they are explicitly engineered into the model (James et al., 2023).

More complex non-linear relationships or interactions may therefore remain uncaptured.

For example, the relationship between price and bestseller performance might differ according to publisher or genre rather than operating as one common linear effect. The current model did not explicitly create such interaction terms.

This does not make Logistic Regression inappropriate for the assessment. Its transparent mathematical structure was particularly suitable for the requirement to develop a classifier from scratch and interpret its coefficients. However, it does constrain the complexity of the relationships the classifier can learn.

11.7 Coefficient and Odds-Ratio Interpretation

The final model used **class-weighted binary cross-entropy**, meaning that bestseller and non-bestseller observations contributed differently to parameter estimation.

For this reason, the resulting coefficients and odds ratios should be treated as relationships within the **fitted weighted predictive model** rather than conventional population-level inferential odds ratios.

Furthermore, because the data are observational, the coefficients cannot demonstrate causal relationships. For example, the positive coefficient associated *with Random House LLC* does not establish that publishing a book through that publisher would cause bestseller performance.

Likewise, the strong negative coefficient for *Author_Rating_Intermediate* relative to the Excellent reference category represents a fitted association after accounting for the other predictors rather than a causal effect of author classification.

11.8 Data-Quality and Encoding Issues

The error-analysis output revealed minor character-encoding artefacts in some book titles. For example, certain non-English titles were displayed using corrupted character sequences.

This represents a data-quality limitation in terms of readability and metadata presentation. However, the impact on the fitted classifier was limited because *Book Name* was excluded

from the predictor matrix due to its high cardinality. The corrupted title text was therefore **not used directly to calculate bestseller probabilities**.

The issue remains relevant when assessing the overall quality of the source dataset and demonstrates why metadata should be inspected rather than assuming that technically non-missing values are necessarily clean.

11.9 Summary of Main Limitations

Limitation	Implication for the analysis
No official bestseller label	Model predicts top-quartile sales within this dataset, not formal industry bestseller status
Limited leakage-safe predictors	Important commercial information was intentionally excluded, reducing predictive signal
Class imbalance	Required class weighting and threshold adjustment
High-recall threshold	Reduced false negatives but generated many false positives
Small categorical groups	Some rates and coefficients are less stable and require cautious interpretation
Single source dataset	External generalisability remains uncertain
Logistic Regression structure	Complex non-linear effects and interactions may remain uncaptured
Weighted predictive model	Odds ratios should not be treated as conventional causal or population-level effects
Minor title-encoding artefacts	Indicates metadata-quality issues, although titles were not model predictors

Table 39: Key limitations affecting interpretation and generalisability of the Logistic Regression analysis

11.10 Section Summary

The main limitations of the analysis arise from the dataset and modelling context rather than from the implementation of Logistic Regression itself. Bestseller status was operationally defined using the top quartile of unit sales rather than an official industry label, while several potentially informative variables had to be excluded to prevent target and temporal leakage. The available target was moderately imbalanced, and the F1-optimised classification threshold produced high recall at the cost of low specificity and numerous false positives. Several categorical groups were small, and the dataset represents only one collection of books rather than the wider publishing market. In addition, Logistic Regression imposes an additive linear structure on the target log-odds. These limitations mean that the final classifier should be interpreted as an exploratory screening model whose results require cautious application rather than as a definitive predictor of commercial success.

12. CONCLUSION

This project developed a **binary Logistic Regression classifier entirely from scratch using Python and NumPy** to investigate whether available book characteristics could be used to identify books with comparatively high sales performance. The analysis covered the complete predictive-modelling process, including dataset assessment, data cleaning, exploratory analysis, feature selection, preprocessing, mathematical model implementation, validation-based improvement, final test evaluation and model interpretation.

The analysis began with **1,070 book records** containing numerical, categorical and commercial characteristics. Because the original dataset did not contain an official bestseller classification, bestseller status was operationally defined using the **75th percentile of *units sold***, equivalent to **5,312.25 units**. This produced 268 bestsellers and 802 non-bestsellers and created a moderately imbalanced binary classification problem.

Careful feature selection was required to ensure that the model represented a realistic predictive problem. Variables such as *units sold, gross sales*, *publisher revenue* and *sales rank* were excluded because they contained realised commercial-performance information that could introduce target leakage. Reader-rating variables were also excluded because they represented information generated after market interaction. The final model therefore relied on six leakage-controlled raw predictors: *Publishing Year*, *sale price*, *Author_Rating*, *genre*, *Publisher* and *language_group*, which expanded to **16 processed model features** after preprocessing.

Exploratory analysis indicated that most available predictors had relatively weak individual relationships with bestseller status. *Author_Rating* produced the clearest categorical association, while sale price and publishing year showed limited class separation. This suggested before modelling that the available characteristics were unlikely to support exceptionally strong discrimination between bestsellers and non-bestsellers.

The Logistic Regression algorithm was then implemented manually rather than through a pre-built estimator. The implementation included the linear predictor, sigmoid transformation, binary cross-entropy loss, analytical gradient calculation, batch gradient descent, probability prediction and threshold-based class prediction. The successful reduction of training loss demonstrated that the custom optimisation procedure was functioning correctly.

The initial model nevertheless revealed the importance of evaluating an imbalanced classifier using more than accuracy. At the conventional threshold of 0.50, the baseline achieved **75.00% validation accuracy**, but predicted every observation as a non-bestseller. Its bestseller recall and F1-score were therefore both zero, giving the same threshold-based performance as a naïve majority-class classifier.

A series of controlled model-improvement experiments was subsequently conducted using validation data only. These included classification-threshold optimisation, balanced class weighting, early stopping and comparison of full and reduced predictor sets. Early stopping retained the weighted model parameters associated with the best validation loss at iteration **618**, while validation-based threshold optimisation selected a final probability threshold of **0.44**.

Although a reduced model containing only *Author_Rating* achieved stronger balanced accuracy and ROC-AUC during validation, the full 16-feature model achieved the slightly higher **F1-score of 0.4186**. Because F1-score had been selected before the comparison as the primary model-selection criterion, the full early-stopped class-weighted model was

retained. This avoided changing the preferred evaluation criterion after observing the model results.

The selected model was finally evaluated once on the untouched test partition. Its performance was:

Metric	**Final test result**
Accuracy	0.3789
Balanced Accuracy	0.5533
Precision	0.2727
Recall	**0.9000**
Specificity	0.2066
F1-Score	**0.4186**
ROC-AUC	**0.6728**

Table 40: Summary of final Logistic Regression test performance

The strongest aspect of the final classifier was its **recall of 0.9000**. It successfully identified **36 of the 40 actual bestsellers**, missing only four. However, this sensitivity came at the cost of a substantial false-positive rate. Of the 121 non-bestsellers, only 25 were correctly rejected, while 96 were incorrectly classified as potential bestsellers. Precision consequently remained low at **0.2727**, and specificity was **0.2066**.

The error analysis supported this interpretation. The four false negatives received predicted probabilities between 0.4208 and 0.4386 and therefore fell narrowly below the 0.44 threshold. In contrast, some false positives received substantially higher probabilities, reaching 0.6389. This showed that the model's limitations could not be explained by threshold selection alone. Instead, considerable overlap existed between the predictor characteristics of bestseller and non-bestseller observations.

Coefficient analysis also supported the earlier exploratory findings. *Author_Rating* produced some of the strongest fitted effects, particularly the Intermediate category relative to the Excellent reference category. Nevertheless, the coefficients should be interpreted as associations within the weighted predictive model rather than causal relationships.

Overall, the project demonstrates that the available pre-outcome book characteristics contain **some useful predictive information**, but they are insufficient for highly reliable bestseller classification. The final model is therefore most appropriately interpreted as an **exploratory high-recall screening tool** that could identify books for further assessment rather than as a system for automatically predicting commercial success.

Most importantly, the project achieved the core technical objective of the assessment by demonstrating the complete development of Logistic Regression **from mathematical formulation through manual gradient-descent optimisation to final evaluation and**

interpretation, while maintaining appropriate separation between training, validation and test data.

12.1 Section Summary

The project successfully implemented and evaluated Logistic Regression from scratch for binary bestseller classification. Controlled model development improved the baseline from a classifier that detected no bestsellers to a final model with **0.9000 recall, 0.4186 F1-score and 0.6728 ROC-AUC** on unseen data. However, low precision and specificity showed that high bestseller sensitivity was achieved at the cost of many false positives. The final results therefore support the use of the model as a preliminary screening aid rather than a definitive commercial forecasting tool.

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